

Theory of Algebraic Topology 16

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Lecture notes with worked examples in algebraic topology

Contents

1 Sphere Fibers and Sphere Bundles

1.1 Basic idea

A *fiber bundle* is a space E , called the *total space*, together with a continuous projection map

$$p : E \rightarrow B,$$

where B is called the *base space*. For each point $b \in B$, the *fiber over b* is the inverse image

$$p^{-1}(b) = \{e \in E : p(e) = b\}.$$

If every fiber is homeomorphic to a sphere S^n , then one obtains an S^n -*bundle*, or *sphere bundle*,

$$S^n \longrightarrow E \xrightarrow{p} B.$$

The geometric meaning is that over every point of the base space B , one places a copy of the sphere S^n . Thus the total space E is built from a whole family of spheres parametrized by the points of B .

Locally, a sphere bundle looks like a product:

$$p^{-1}(U) \cong U \times S^n,$$

where $U \subseteq B$ is a sufficiently small open set. Globally, however, the bundle may be twisted, so in general one need not have

$$E \cong B \times S^n.$$

This is the main idea of a fiber bundle: locally it is simple, but globally it may contain nontrivial twisting.

1.2 The bundle diagram

A sphere bundle is often represented by the diagram

$$\begin{array}{ccc} S^n & \longrightarrow & E \\ & & \downarrow p \\ & & B \end{array}$$

Here S^n is the typical fiber, E is the total space, and $p : E \rightarrow B$ is the projection map. For every point $b \in B$, the fiber is

$$p^{-1}(b) \cong S^n.$$

1.3 Local triviality

The defining condition of a fiber bundle is *local triviality*. This means that for every point $b \in B$, there exists an open neighborhood $U \subseteq B$ such that the part of the total space lying above U is homeomorphic to a product:

$$p^{-1}(U) \cong U \times S^n.$$

This homeomorphism is required to respect the projection to U . In diagram form, this means that the following diagram commutes:

$$\begin{array}{ccc} p^{-1}(U) & \xrightarrow{\varphi_U} & U \times S^n \\ p \downarrow & & \downarrow \text{pr}_1 \\ U & \xlongequal{\quad} & U. \end{array}$$

Thus, over a small open set U , the bundle behaves exactly like the product $U \times S^n$. The nontriviality of the bundle comes from how these local product pieces are glued together on overlaps.

1.4 Transition functions and twisting

Suppose B is covered by open sets $\{U_i\}$, and over each U_i the bundle is trivial:

$$p^{-1}(U_i) \cong U_i \times S^n.$$

On an overlap $U_i \cap U_j$, the two local trivializations may differ by a map

$$g_{ij} : U_i \cap U_j \rightarrow \text{Homeo}(S^n).$$

These maps are called *transition functions*. They describe how the spherical fibers are glued together.

For sphere bundles coming from real vector bundles, the transition functions often lie in an orthogonal group:

$$g_{ij} : U_i \cap U_j \rightarrow O(n+1),$$

because S^n is the unit sphere in $\mathbb{R}^{n+1}$.

Thus one can think of a sphere bundle as a product $B \times S^n$ locally, but with possible rotations, reflections, or more general transformations used to glue the fibers together globally.

1.5 Sphere bundles from vector bundles

One of the most important sources of sphere bundles comes from vector bundles.

Let

$$\pi : V \rightarrow B$$

be a real vector bundle of rank r . This means that each fiber is a vector space

$$V_b \cong \mathbb{R}^r.$$

If the vector bundle is equipped with a fiberwise norm, then inside each vector space V_b we can take the unit sphere

$$S(V_b) = \{v \in V_b : \|v\| = 1\}.$$

The union of all these unit spheres forms the *unit sphere bundle*

$$S(V) = \{v \in V : \|v\| = 1\}.$$

The projection

$$S(V) \rightarrow B$$

has fiber

$$S^{r-1}.$$

Therefore a rank r vector bundle gives rise to an S^{r-1} -bundle:

rank r vector bundle $V \rightarrow B \implies S^{r-1}$ -bundle $S(V) \rightarrow B.$

For example, if $V \rightarrow B$ has rank 3, then every fiber is isomorphic to $\mathbb{R}^3$, and the unit sphere in each fiber is

$$S^2 \subset \mathbb{R}^3.$$

Hence the associated unit sphere bundle is an S^2 -bundle over B .

1.6 Disk bundles and sphere bundles

From a vector bundle $V \rightarrow B$, one can also form the *unit disk bundle*

$$D(V) = \{v \in V : \|v\| \leq 1\}.$$

Each fiber is a disk:

$$D(V)_b \cong D^r.$$

The boundary of this disk fiber is the unit sphere:

$$\partial D^r = S^{r-1}.$$

Therefore, fiber by fiber, the boundary of the disk bundle is the sphere bundle:

$\partial D(V) = S(V).$

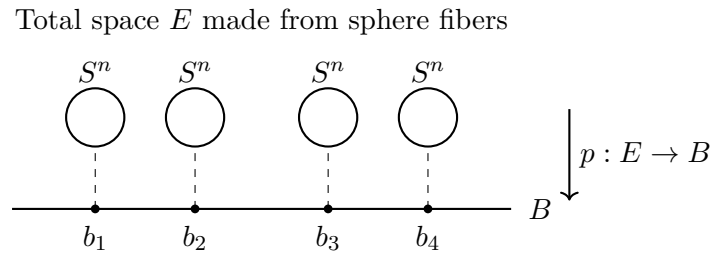
Thus disk bundles and sphere bundles are closely related:

$$S(V) \subset D(V) \subset V.$$

In each fiber, this is simply the inclusion

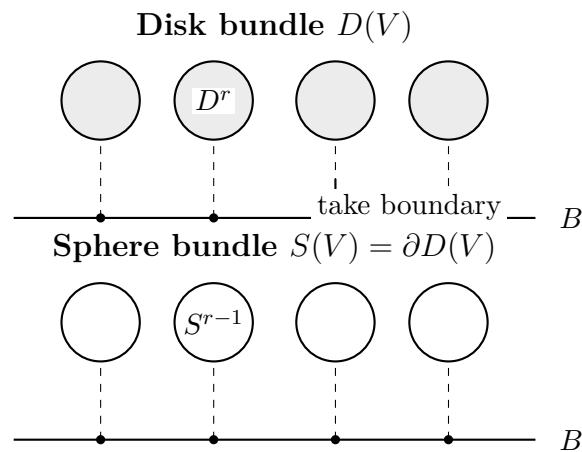
$$S^{r-1} \subset D^r \subset \mathbb{R}^r.$$

1.7 Diagram: general sphere bundle



The picture shows that every point $b \in B$ has a sphere S^n lying above it. The total space E is obtained by collecting all these fibers.

1.8 Diagram: disk bundle and sphere boundary



This diagram expresses the fiberwise identity

$$\partial D^r = S^{r-1},$$

and therefore

$$\partial D(V) = S(V).$$

1.9 Example 1: The trivial sphere bundle

The simplest sphere bundle is the product bundle

$$E = B \times S^n.$$

The projection is

$$p : B \times S^n \rightarrow B, \quad p(b, x) = b.$$

The fiber over a point $b \in B$ is

$$p^{-1}(b) = \{b\} \times S^n \cong S^n.$$

This bundle is called *trivial* because the total space is globally a product:

$$E \cong B \times S^n.$$

There is no twisting.

1.10 Example 2: The cylinder as an S^1 -bundle

Consider

$$E = [0, 1] \times S^1.$$

The projection

$$p : [0, 1] \times S^1 \rightarrow [0, 1], \quad p(t, z) = t,$$

makes E into an S^1 -bundle over the interval $[0, 1]$.

For each point $t \in [0, 1]$, the fiber is

$$p^{-1}(t) = \{t\} \times S^1 \cong S^1.$$

Geometrically, this is a cylinder. The base is an interval, and above every point of the interval sits a circle.

1.11 Example 3: The torus as a trivial circle bundle

The torus can be written as a product

$$T^2 = S^1 \times S^1.$$

Projection onto the first factor gives

$$p : S^1 \times S^1 \rightarrow S^1, \quad p(x, y) = x.$$

The fiber over $x \in S^1$ is

$$p^{-1}(x) = \{x\} \times S^1 \cong S^1.$$

Thus

$$S^1 \longrightarrow T^2 \longrightarrow S^1$$

is a trivial circle bundle.

In this case, the base is a circle and the fiber is also a circle. Since there is no twisting, the total space is the product of two circles:

$$T^2 = S^1 \times S^1.$$

1.12 Example 4: The Klein bottle as a nontrivial circle bundle

The Klein bottle is also an S^1 -bundle over S^1 :

$$S^1 \longrightarrow K \longrightarrow S^1.$$

However, unlike the torus, the Klein bottle is not the product $S^1 \times S^1$. When one travels once around the base circle, the circle fiber is reflected.

Thus the torus and Klein bottle can be compared as follows:

$$\boxed{T^2 = S^1 \times S^1 \text{ is the trivial circle bundle over } S^1.}$$

$$\boxed{K \text{ is the nontrivial circle bundle over } S^1.}$$

The difference is the twisting of the fiber. The torus glues circle fibers without reflection, while the Klein bottle glues them with a reflection.

1.13 Example 5: The Hopf fibration

One of the most important examples of a sphere bundle is the *Hopf fibration*

$$S^1 \longrightarrow S^3 \longrightarrow S^2.$$

Here the total space is S^3 , the base is S^2 , and the fiber is S^1 .

This means that over every point of S^2 , there is a circle S^1 . The total space obtained from all these circle fibers is S^3 .

The Hopf fibration is not a product:

$$S^3 \not\cong S^2 \times S^1.$$

Indeed, the topology is different. For example,

$$\pi_1(S^3) = 0,$$

whereas

$$\pi_1(S^2 \times S^1) \cong \mathbb{Z}.$$

Thus the Hopf fibration is a genuinely nontrivial circle bundle.

Geometrically, the circle fibers in the Hopf fibration are linked with one another inside S^3 . This linking behavior is one of the most famous features of the Hopf fibration.

1.14 Example 6: The unit tangent bundle of S^2

For a smooth manifold M , the tangent bundle is

$$TM \rightarrow M.$$

The fiber over a point $x \in M$ is the tangent space

$$T_x M.$$

If $M = S^2$, then each tangent space is two-dimensional:

$$T_x S^2 \cong \mathbb{R}^2.$$

The unit vectors in $T_x S^2$ form a circle:

$$S(T_x S^2) \cong S^1.$$

Therefore the unit tangent bundle of the sphere,

$$US^2 \rightarrow S^2,$$

is an S^1 -bundle:

$$S^1 \longrightarrow US^2 \longrightarrow S^2.$$

This bundle is nontrivial. In fact, one has the important identification

$$US^2 \cong SO(3) \cong \mathbb{RP}^3.$$

This example shows that even very natural sphere bundles, such as the unit tangent bundle of a sphere, can be globally nontrivial.

1.15 Summary table of examples

Construction	Base B	Fiber	Total space E	Trivial?
Product bundle	B	S^n	$B \times S^n$	Yes
Cylinder	$[0, 1]$	S^1	$[0, 1] \times S^1$	Yes
Torus projection	S^1	S^1	$T^2 = S^1 \times S^1$	Yes
Klein bottle	S^1	S^1	K	No
Hopf fibration	S^2	S^1	S^3	No
Unit tangent bundle of S^2	S^2	S^1	$US^2 \cong \mathbb{RP}^3$	No
Unit sphere bundle of rank r vector bundle	B	S^{r-1}	$S(V)$	Depends

1.16 Chart: vector fiber, disk fiber, and sphere fiber

Fiber object	Shape	Formula	Boundary
Vector fiber	$\mathbb{R}^r$	all vectors v	none
Disk fiber	D^r	$\ v\ \leq 1$	S^{r-1}
Sphere fiber	S^{r-1}	$\ v\ = 1$	none

Thus, inside each vector fiber, one has

$$S^{r-1} \subset D^r \subset \mathbb{R}^r.$$

Fiberwise over the base B , this becomes

$$S(V) \subset D(V) \subset V.$$

1.17 Conceptual comparison: trivial and nontrivial bundles

Feature	Trivial sphere bundle	Nontrivial sphere bundle
Local form	$U \times S^n$	$U \times S^n$
Global form	$B \times S^n$	not necessarily $B \times S^n$
Twisting	no twisting	fibers twist when glued
Example	$T^2 = S^1 \times S^1$	Klein bottle, Hopf fibration
Transition functions	can be chosen globally constant	carry nontrivial gluing information

The important point is that both trivial and nontrivial bundles are locally products. The difference is global.

1.18 Geometric intuition

A sphere bundle is not a single sphere. It is a family of spheres indexed by the base space.

For each point $b \in B$, there is a sphere above it:

$$b \longmapsto p^{-1}(b) \cong S^n.$$

The total space is the union of all these fibers:

$$E = \bigcup_{b \in B} p^{-1}(b).$$

If the fibers are glued together without twisting, then the total space is a product:

$$E \cong B \times S^n.$$

If the fibers are glued together with twisting, the total space is a nontrivial sphere bundle.

Therefore the main intuition is:

A sphere bundle measures how spherical fibers twist over a base space.

Sphere bundles appear naturally in topology, differential geometry, vector bundle theory, characteristic classes, tangent bundles, normal bundles, and fibrations such as the Hopf fibration.

2 Problems 9–11: Vector Bundles, Euler Characteristic, and Lefschetz Fixed Points

Throughout this section, manifolds are assumed to be compact when necessary, and cohomology is taken with coefficients in a field F when the Lefschetz pairing is discussed.

3 Problem 9

Problem. Construct a 3-dimensional real vector bundle over S^4 which has no nonvanishing section.

Solution.

We construct the bundle by the clutching construction.

Write

$$S^4 = D_+^4 \cup D_-^4,$$

where

$$D_+^4 \cap D_-^4 = S^3.$$

A real oriented rank-3 vector bundle over S^4 is obtained by taking two trivial bundles

$$D_+^4 \times \mathbb{R}^3 \quad \text{and} \quad D_-^4 \times \mathbb{R}^3$$

and gluing them along the equator S^3 by a map

$$g : S^3 \rightarrow SO(3).$$

Such maps are classified up to homotopy by

$$\pi_3(SO(3)).$$

It is known that

$$\pi_3(SO(3)) \cong \mathbb{Z}.$$

Choose $g : S^3 \rightarrow SO(3)$ to represent a nonzero element of

$$\pi_3(SO(3)).$$

For example, one may take the universal covering map

$$S^3 \cong SU(2) \longrightarrow SO(3).$$

Let

$$E_g \rightarrow S^4$$

be the rank-3 real vector bundle obtained from this nontrivial clutching map.

We claim that E_g has no nonvanishing section.

Suppose, for contradiction, that E_g had a nowhere-zero section

$$s : S^4 \rightarrow E_g.$$

After choosing a bundle metric, we can normalize s and assume that it is a unit section. Then each fiber splits as

$$(E_g)_x = \langle s(x) \rangle \oplus s(x)^\perp.$$

Therefore the bundle splits as

$$E_g \cong \varepsilon^1 \oplus \xi,$$

where ε^1 is the trivial real line bundle and ξ is an oriented real 2-plane bundle.

Equivalently, the structure group of E_g reduces from $SO(3)$ to the stabilizer of a unit vector in $\mathbb{R}^3$. This stabilizer is

$$SO(2) \subset SO(3).$$

Thus the clutching map $g : S^3 \rightarrow SO(3)$ would be homotopic to a map whose image lies in $SO(2)$. In other words, its homotopy class would lie in the image of

$$\pi_3(SO(2)) \longrightarrow \pi_3(SO(3)).$$

But

$$SO(2) \cong S^1,$$

and hence

$$\pi_3(SO(2)) = \pi_3(S^1) = 0.$$

Therefore the image of

$$\pi_3(SO(2)) \rightarrow \pi_3(SO(3))$$

is zero. This would force the clutching class of E_g in

$$\pi_3(SO(3)) \cong \mathbb{Z}$$

to be zero, contradicting the choice of g .

Hence E_g has no nonvanishing section.

Therefore

A rank-3 real vector bundle over S^4 defined by a nonzero clutching map $S^3 \rightarrow SO(3)$ has no nonvanishing section.

4 Problem 10

Problem. If M is a closed odd-dimensional manifold, show that

$$\chi(M) = 0.$$

Next, suppose M is a compact odd-dimensional manifold with boundary. Show that

$$\chi(M) = \frac{1}{2}\chi(\partial M).$$

Conclude that $\mathbb{RP}^2$ does not bound a compact 3-manifold and that $\mathbb{CP}^2$ does not bound a compact 5-manifold. Does $\mathbb{RP}^3$ bound a compact 4-manifold? Does $\mathbb{CP}^3$ bound a compact 7-manifold?

Solution.

Step 1: Closed odd-dimensional manifolds

Let M be a closed n -dimensional manifold, where n is odd. We compute the Euler characteristic using Betti numbers over a field F :

$$\chi(M) = \sum_{i=0}^n (-1)^i b_i(M),$$

where

$$b_i(M) = \dim_F H_i(M; F).$$

By Poincaré duality,

$$H_i(M; F) \cong H_{n-i}(M; F),$$

so

$$b_i(M) = b_{n-i}(M).$$

Since n is odd, the terms in the Euler characteristic cancel in pairs:

$$(-1)^i b_i + (-1)^{n-i} b_{n-i} = (-1)^i b_i + (-1)^{n-i} b_i.$$

But n is odd, so

$$(-1)^{n-i} = -(-1)^i.$$

Therefore

$$(-1)^i b_i + (-1)^{n-i} b_i = 0.$$

All terms cancel pairwise, and hence

$$\chi(M) = 0.$$

Thus

$\chi(M) = 0 \quad \text{for every closed odd-dimensional manifold } M.$

Step 2: Compact odd-dimensional manifolds with boundary

Now suppose M is a compact odd-dimensional manifold with boundary

$$\partial M.$$

Consider the double of M :

$$DM = M \cup_{\partial M} M.$$

The double DM is a closed odd-dimensional manifold. Therefore, by Step 1,

$$\chi(DM) = 0.$$

Using additivity of Euler characteristic, we get

$$\chi(DM) = \chi(M) + \chi(M) - \chi(\partial M).$$

Hence

$$0 = 2\chi(M) - \chi(\partial M).$$

Therefore

$$\chi(M) = \frac{1}{2}\chi(\partial M).$$

Thus

$$\boxed{\chi(M) = \frac{1}{2}\chi(\partial M)}$$

for every compact odd-dimensional manifold M with boundary.

Step 3: Consequence for boundaries

If a closed manifold N bounds a compact odd-dimensional manifold M , then

$$N = \partial M.$$

The formula above gives

$$\chi(M) = \frac{1}{2}\chi(N).$$

Since $\chi(M)$ is an integer, $\chi(N)$ must be even.

Thus a necessary condition for N to bound a compact odd-dimensional manifold is:

$$\chi(N) \equiv 0 \pmod{2}.$$

Step 4: $\mathbb{RP}^2$

The Euler characteristic of real projective space is

$$\chi(\mathbb{RP}^2) = 1.$$

This is odd. Therefore $\mathbb{RP}^2$ cannot be the boundary of a compact 3-manifold.

Hence

$$\boxed{\mathbb{RP}^2 \text{ does not bound a compact 3-manifold.}}$$

Step 5: $\mathbb{CP}^2$

The Euler characteristic of complex projective space is

$$\chi(\mathbb{CP}^n) = n + 1.$$

Thus

$$\chi(\mathbb{CP}^2) = 3.$$

This is odd. Therefore $\mathbb{CP}^2$ cannot be the boundary of a compact 5-manifold.

Hence

$$\boxed{\mathbb{CP}^2 \text{ does not bound a compact 5-manifold.}}$$

Step 6: Does $\mathbb{RP}^3$ bound a compact 4-manifold?

Yes.

A standard construction uses the disk bundle of an oriented real 2-plane bundle over S^2 with Euler number 2. Let

$$\xi \rightarrow S^2$$

be an oriented real 2-plane bundle with Euler class

$$e(\xi) = 2 \in H^2(S^2; \mathbb{Z}) \cong \mathbb{Z}.$$

Let

$$D(\xi)$$

be its disk bundle and

$$S(\xi)$$

its circle bundle. Then

$$\partial D(\xi) = S(\xi).$$

The oriented circle bundle over S^2 with Euler number 2 is the lens space

$$L(2, 1),$$

and

$$L(2, 1) \cong \mathbb{RP}^3.$$

Hence

$$\partial D(\xi) \cong \mathbb{RP}^3.$$

Therefore

$\mathbb{RP}^3$ bounds a compact 4-manifold.

Step 7: Does $\mathbb{CP}^3$ bound a compact 7-manifold?

Yes.

Recall that $\mathbb{CP}^3$ may be realized as the twistor space of S^4 . Equivalently,

$$\mathbb{CP}^3 \cong S(\Lambda_+^2(TS^4)),$$

where

$$\Lambda_+^2(TS^4) \rightarrow S^4$$

is the real rank-3 vector bundle of self-dual 2-forms on S^4 , and $S(\Lambda_+^2(TS^4))$ denotes its sphere bundle.

Let

$$D(\Lambda_+^2(TS^4))$$

be the corresponding disk bundle. Since a sphere bundle is the boundary of the disk bundle, we have

$$\partial D(\Lambda_+^2(TS^4)) = S(\Lambda_+^2(TS^4)).$$

Therefore

$$\partial D(\Lambda_+^2(TS^4)) \cong \mathbb{CP}^3.$$

The base S^4 has dimension 4, and the disk fibers have dimension 3. Hence the disk bundle has dimension

$$4 + 3 = 7.$$

Therefore

$$\boxed{\mathbb{CP}^3 \text{ bounds a compact 7-manifold.}}$$

5 Problem 11

Problem. Suppose F is a field, and M is a closed F -oriented manifold. Let

$$\{a_i\}$$

be a homogeneous basis of $H^*(M; F)$, and let

$$\{b_i\}$$

be the dual basis with respect to the cup product pairing, meaning

$$\langle a_i \smile b_j, [M] \rangle = \delta_{ij}.$$

Given a map

$$f : M \rightarrow M,$$

let

$$\Lambda_f = \{(p, f(p)) \in M \times M\}$$

be the graph of f .

(a) Express $PD_{M \times M}(\Lambda_f)$ in terms of the a_i and b_i .

(b) Define the Lefschetz number of f by

$$L(f) = \sum_{j=0}^n (-1)^j \operatorname{Tr} f_j^*,$$

where

$$f_j^* : H^j(M; F) \rightarrow H^j(M; F).$$

Show that

$$L(f) = \Lambda_f \cdot \Delta,$$

where $\Delta \subset M \times M$ is the diagonal.

(c) Deduce that if $L(f) \neq 0$, then f must have a fixed point.

(d) Show that any map

$$f : \mathbb{CP}^2 \rightarrow \mathbb{CP}^2$$

has a fixed point.

Solution.

Part (a)

Let

$$\Delta = \{(p, p) \in M \times M\}$$

be the diagonal. The standard formula for the Poincaré dual of the diagonal is

$$PD_{M \times M}(\Delta) = \sum_i (-1)^{|a_i|} p_1^*(a_i) \smile p_2^*(b_i),$$

where

$$p_1, p_2 : M \times M \rightarrow M$$

are the projections, and $|a_i|$ denotes the degree of a_i .

The graph of f is

$$\Lambda_f = \{(p, f(p)) \mid p \in M\}.$$

It is the image of the embedding

$$\Gamma_f : M \rightarrow M \times M, \quad \Gamma_f(p) = (p, f(p)).$$

Equivalently, Λ_f is obtained from the diagonal by applying the map

$$\text{id} \times f : M \times M \rightarrow M \times M$$

to the first copy of M in the diagonal picture. The corresponding Poincaré dual formula is obtained by applying f^* to the second factor in the diagonal formula. Thus

$$PD_{M \times M}(\Lambda_f) = \sum_i (-1)^{|a_i|} p_1^*(a_i) \smile p_2^*(f^*b_i).$$

Equivalently, depending on the convention for the graph

$$p \longmapsto (f(p), p)$$

rather than

$$p \longmapsto (p, f(p)),$$

one obtains the equivalent formula

$$PD_{M \times M}(\Lambda_f) = \sum_i (-1)^{|a_i|} p_1^*(f^*a_i) \smile p_2^*(b_i).$$

With the convention in this problem,

$$\Lambda_f = \{(p, f(p))\},$$

we use

$$\boxed{PD_{M \times M}(\Lambda_f) = \sum_i (-1)^{|a_i|} p_1^*(a_i) \smile p_2^*(f^*b_i).}$$

Part (b)

We want to show that

$$L(f) = \Lambda_f \cdot \Delta.$$

The intersection number of Λ_f with the diagonal Δ is

$$\Lambda_f \cdot \Delta = \langle PD(\Lambda_f) \smile PD(\Delta), [M \times M] \rangle.$$

There is a useful equivalent way to compute this intersection number. Pull back $PD(\Lambda_f)$ along the diagonal map

$$\delta : M \rightarrow M \times M, \quad \delta(p) = (p, p).$$

Then

$$\Lambda_f \cdot \Delta = \langle \delta^* PD(\Lambda_f), [M] \rangle.$$

Using the formula from part (a),

$$PD(\Lambda_f) = \sum_i (-1)^{|a_i|} p_1^*(a_i) \smile p_2^*(f^*b_i).$$

Since

$$p_1 \circ \delta = \text{id}_M, \quad p_2 \circ \delta = \text{id}_M,$$

we get

$$\delta^* PD(\Lambda_f) = \sum_i (-1)^{|a_i|} a_i \smile f^*b_i.$$

Therefore

$$\Lambda_f \cdot \Delta = \sum_i (-1)^{|a_i|} \langle a_i \smile f^*b_i, [M] \rangle.$$

Now group the basis elements by degree. For each j , let

$$\{a_{j,\alpha}\}_\alpha$$

be a basis of $H^j(M; F)$, and let

$$\{b_{j,\alpha}\}_\alpha$$

be the dual basis in $H^{n-j}(M; F)$, so that

$$\langle a_{j,\alpha} \smile b_{j,\beta}, [M] \rangle = \delta_{\alpha\beta}.$$

The map

$$f^* : H^{n-j}(M; F) \rightarrow H^{n-j}(M; F)$$

has a matrix in the basis $\{b_{j,\alpha}\}$. By duality, this trace agrees with the trace of

$$f^* : H^j(M; F) \rightarrow H^j(M; F).$$

Therefore

$$\sum_\alpha \langle a_{j,\alpha} \smile f^*b_{j,\alpha}, [M] \rangle = \text{Tr} \left(f^* : H^j(M; F) \rightarrow H^j(M; F) \right).$$

Since $|a_{j,\alpha}| = j$, we obtain

$$\Lambda_f \cdot \Delta = \sum_{j=0}^n (-1)^j \operatorname{Tr} \left(f^* : H^j(M; F) \rightarrow H^j(M; F) \right).$$

By definition, the right-hand side is the Lefschetz number:

$$L(f) = \sum_{j=0}^n (-1)^j \operatorname{Tr}(f_j^*).$$

Hence

$$\boxed{L(f) = \Lambda_f \cdot \Delta.}$$

Part (c)

A point of the graph Λ_f has the form

$$(p, f(p)).$$

A point of the diagonal Δ has the form

$$(p, p).$$

Therefore

$$(p, f(p)) \in \Delta$$

if and only if

$$f(p) = p.$$

Thus

$$\Lambda_f \cap \Delta$$

is precisely the set of fixed points of f .

If f had no fixed point, then

$$\Lambda_f \cap \Delta = \emptyset.$$

In that case their intersection number would be zero:

$$\Lambda_f \cdot \Delta = 0.$$

But by part (b),

$$\Lambda_f \cdot \Delta = L(f).$$

Hence if f has no fixed point, then

$$L(f) = 0.$$

Taking the contrapositive, if

$$L(f) \neq 0,$$

then f must have a fixed point.

Therefore

$$\boxed{L(f) \neq 0 \implies f \text{ has a fixed point.}}$$

This is the Lefschetz fixed point theorem in this setting.

Part (d)

Let

$$f : \mathbb{CP}^2 \rightarrow \mathbb{CP}^2$$

be any continuous map. We will show that f has a fixed point.

Work with rational coefficients:

$$F = \mathbb{Q}.$$

The cohomology ring of $\mathbb{CP}^2$ is

$$H^*(\mathbb{CP}^2; \mathbb{Q}) \cong \mathbb{Q}[h]/(h^3), \quad \deg h = 2.$$

Thus

$$\begin{aligned} H^0(\mathbb{CP}^2; \mathbb{Q}) &\cong \mathbb{Q}, \\ H^2(\mathbb{CP}^2; \mathbb{Q}) &\cong \mathbb{Q}\langle h \rangle, \end{aligned}$$

and

$$H^4(\mathbb{CP}^2; \mathbb{Q}) \cong \mathbb{Q}\langle h^2 \rangle.$$

All other cohomology groups vanish.

Since $H^2(\mathbb{CP}^2; \mathbb{Q}) \cong \mathbb{Q}$, the map f^* acts by multiplication by some integer d :

$$f^*(h) = d h.$$

Because f^* preserves cup products,

$$f^*(h^2) = f^*(h \smile h) = f^*(h) \smile f^*(h) = d^2 h^2.$$

Therefore the traces of f^* on cohomology are:

$$\begin{aligned} \text{Tr}(f^* : H^0 \rightarrow H^0) &= 1, \\ \text{Tr}(f^* : H^2 \rightarrow H^2) &= d, \end{aligned}$$

and

$$\text{Tr}(f^* : H^4 \rightarrow H^4) = d^2.$$

Since all nonzero cohomology groups occur in even degrees, the Lefschetz number is

$$L(f) = 1 + d + d^2.$$

Now

$$d^2 + d + 1$$

is never zero for an integer d . Indeed,

$$d^2 + d + 1 = \left(d + \frac{1}{2}\right)^2 + \frac{3}{4} > 0.$$

Therefore

$$L(f) \neq 0.$$

By part (c), f must have a fixed point.

Hence

Every continuous map $f : \mathbb{CP}^2 \rightarrow \mathbb{CP}^2$ has a fixed point.

6 Theory for Problems 9–11: Clutching, Euler Characteristic, Boundaries, and Lefschetz Fixed Points

These three problems are connected by a general theme:

$$\text{global topology} \longrightarrow \text{algebraic invariant} \longrightarrow \text{geometric consequence}.$$

Problem 9 uses clutching functions to construct a vector bundle with no nonzero section. Problem 10 uses Poincaré duality and Euler characteristic to obstruct some manifolds from being boundaries. Problem 11 uses the Lefschetz number as an intersection number between the graph of a map and the diagonal.

6.1 Vector Bundles over Spheres and Clutching Functions

Let

$$E \rightarrow S^n$$

be a rank- r real vector bundle. We can decompose the sphere as two disks glued along their common boundary:

$$S^n = D_+^n \cup D_-^n,$$

with

$$D_+^n \cap D_-^n = S^{n-1}.$$

Since each disk is contractible, every vector bundle over each disk is trivial:

$$E|_{D_+^n} \cong D_+^n \times \mathbb{R}^r,$$

and

$$E|_{D_-^n} \cong D_-^n \times \mathbb{R}^r.$$

Thus the bundle is determined by how these two trivial bundles are glued together along the equator S^{n-1} . The gluing is described by a map

$$g : S^{n-1} \rightarrow GL_r(\mathbb{R}).$$

If the bundle is oriented and equipped with a metric, we may take

$$g : S^{n-1} \rightarrow SO(r).$$

This map is called the *clutching function*. The homotopy class of g determines the isomorphism class of the vector bundle. For oriented rank- r real vector bundles over S^n , one obtains the classification

$$\{\text{oriented rank-}r \text{ bundles over } S^n\} \cong \pi_{n-1}(SO(r)).$$

6.2 Rank-3 Bundles over S^4

In Problem 9, we want a rank-3 real vector bundle over S^4 . The relevant clutching group is

$$\pi_3(SO(3)).$$

A classical fact is

$$\pi_3(SO(3)) \cong \mathbb{Z}.$$

Therefore there are infinitely many oriented rank-3 real vector bundles over S^4 . The trivial bundle corresponds to the zero element of

$$\pi_3(SO(3)).$$

A nontrivial clutching map

$$g : S^3 \rightarrow SO(3)$$

produces a nontrivial rank-3 bundle.

A standard nontrivial example comes from the double covering

$$SU(2) \cong S^3 \longrightarrow SO(3).$$

This map represents a nonzero element of $\pi_3(SO(3))$.

6.3 Nonvanishing Sections and Reduction of Structure Group

Let

$$E \rightarrow B$$

be an oriented rank-3 real vector bundle.

A nowhere-zero section

$$s : B \rightarrow E$$

chooses a nonzero vector in every fiber. After choosing a bundle metric, we can normalize it and assume

$$\|s(b)\| = 1$$

for every $b \in B$.

Then every fiber splits as

$$E_b = \langle s(b) \rangle \oplus s(b)^\perp.$$

Globally, this gives a splitting

$$E \cong \varepsilon^1 \oplus \xi,$$

where ε^1 is the trivial real line bundle and ξ is an oriented rank-2 real vector bundle.

Equivalently, the structure group reduces from

$$SO(3)$$

to the stabilizer of a unit vector in $\mathbb{R}^3$. This stabilizer is

$$SO(2).$$

Thus

a nonvanishing section $\implies$ the clutching map factors through $SO(2)$.

For a bundle over S^4 , this means that the clutching map would come from a map

$$S^3 \rightarrow SO(2).$$

But

$$SO(2) \cong S^1,$$

and

$$\pi_3(S^1) = 0.$$

Therefore every $SO(2)$ -valued clutching map over S^4 is null-homotopic. So a rank-3 bundle over S^4 with nonzero clutching class in

$$\pi_3(SO(3)) \cong \mathbb{Z}$$

cannot admit a nowhere-zero section.

This is the main idea behind Problem 9.

6.4 Why the Euler Class Alone Does Not Solve Problem 9

For an oriented rank-3 bundle

$$E \rightarrow S^4,$$

the Euler class would live in

$$e(E) \in H^3(S^4; \mathbb{Z}).$$

But

$$H^3(S^4; \mathbb{Z}) = 0.$$

Therefore

$$e(E) = 0$$

for every oriented rank-3 bundle over S^4 .

So the Euler class cannot detect the obstruction in Problem 9. The obstruction is instead detected by the clutching map:

$$[g] \in \pi_3(SO(3)).$$

This is an important lesson: the Euler class is not the only obstruction to a nonzero section. In this example, the obstruction is homotopical rather than cohomological in degree 3.

6.5 Euler Characteristic

For a finite CW complex or compact manifold M , the Euler characteristic is

$$\chi(M) = \sum_{i=0}^n (-1)^i b_i(M),$$

where

$$b_i(M) = \dim_F H_i(M; F)$$

is the i -th Betti number over a field F .

The Euler characteristic is additive under decompositions. If

$$X = A \cup B,$$

then

$$\chi(X) = \chi(A) + \chi(B) - \chi(A \cap B).$$

This additivity is crucial in the boundary formula from Problem 10.

6.6 Closed Odd-Dimensional Manifolds Have Euler Characteristic Zero

Let M^n be a closed manifold with n odd. Poincaré duality over a field gives

$$H_i(M; F) \cong H_{n-i}(M; F).$$

Therefore

$$b_i = b_{n-i}.$$

Now compute

$$\chi(M) = \sum_{i=0}^n (-1)^i b_i.$$

Pair the i -th term with the $(n-i)$ -th term:

$$(-1)^i b_i + (-1)^{n-i} b_{n-i}.$$

Since $b_i = b_{n-i}$, this becomes

$$\left((-1)^i + (-1)^{n-i} \right) b_i.$$

Because n is odd,

$$(-1)^{n-i} = -(-1)^i.$$

Thus each pair cancels:

$$(-1)^i b_i + (-1)^{n-i} b_{n-i} = 0.$$

Hence

$$\chi(M) = 0.$$

Therefore

Every closed odd-dimensional manifold has Euler characteristic 0.

6.7 Compact Odd-Dimensional Manifolds with Boundary

Now suppose M is compact, odd-dimensional, and has boundary

$$\partial M.$$

The key construction is the *double* of M :

$$DM = M \cup_{\partial M} M.$$

This means that we glue two copies of M along their common boundary. The double DM is a closed odd-dimensional manifold. Therefore

$$\chi(DM) = 0.$$

Using additivity of Euler characteristic, we get

$$\chi(DM) = \chi(M) + \chi(M) - \chi(\partial M).$$

Hence

$$0 = 2\chi(M) - \chi(\partial M).$$

Therefore

$$\chi(M) = \frac{1}{2}\chi(\partial M).$$

Thus

$$\boxed{\chi(M) = \frac{1}{2}\chi(\partial M)}$$

for every compact odd-dimensional manifold M with boundary.

6.8 Euler Characteristic as a Boundary Obstruction

Suppose N is a closed even-dimensional manifold and

$$N = \partial M$$

for some compact odd-dimensional manifold M .

Then

$$\chi(M) = \frac{1}{2}\chi(N).$$

Since $\chi(M)$ is an integer, we must have

$$\chi(N) \in 2\mathbb{Z}.$$

Thus

$$\boxed{\text{If } N \text{ bounds a compact odd-dimensional manifold, then } \chi(N) \text{ is even.}}$$

This gives a simple obstruction to being a boundary.

6.9 Real and Complex Projective Spaces

The Euler characteristics needed in Problem 10 are

$$\chi(\mathbb{RP}^{2m}) = 1,$$

$$\chi(\mathbb{RP}^{2m+1}) = 0,$$

and

$$\chi(\mathbb{CP}^n) = n + 1.$$

Thus

$$\chi(\mathbb{RP}^2) = 1,$$

$$\chi(\mathbb{RP}^3) = 0,$$

$$\chi(\mathbb{CP}^2) = 3,$$

and

$$\chi(\mathbb{CP}^3) = 4.$$

Therefore $\mathbb{RP}^2$ cannot bound a compact 3-manifold, because

$$\chi(\mathbb{RP}^2) = 1$$

is odd.

Similarly, $\mathbb{CP}^2$ cannot bound a compact 5-manifold, because

$$\chi(\mathbb{CP}^2) = 3$$

is odd.

On the other hand,

$$\chi(\mathbb{RP}^3) = 0$$

and

$$\chi(\mathbb{CP}^3) = 4$$

are even. Hence Euler characteristic does not obstruct them from bounding.

Indeed, both do bound compact manifolds in the examples described below.

6.10 Examples of Boundaries

$\mathbb{RP}^3$

The space

$$\mathbb{RP}^3$$

is diffeomorphic to the lens space

$$L(2, 1).$$

It is the circle bundle over S^2 with Euler number 2. Equivalently, if

$$\xi \rightarrow S^2$$

is the oriented real 2-plane bundle with

$$e(\xi) = 2 \in H^2(S^2; \mathbb{Z}) \cong \mathbb{Z},$$

then its disk bundle $D(\xi)$ is a compact 4-manifold and

$$\partial D(\xi) = S(\xi) \cong \mathbb{RP}^3.$$

Thus

$$\boxed{\mathbb{RP}^3 \text{ bounds a compact 4-manifold.}}$$

$\mathbb{CP}^3$

The space

$$\mathbb{CP}^3$$

can be realized as the twistor space of S^4 . Equivalently,

$$\mathbb{CP}^3 \cong S(\Lambda_+^2(TS^4)),$$

where

$$\Lambda_+^2(TS^4) \rightarrow S^4$$

is the rank-3 real vector bundle of self-dual 2-forms.

The sphere bundle is the boundary of the disk bundle:

$$S(\Lambda_+^2(TS^4)) = \partial D(\Lambda_+^2(TS^4)).$$

Since the base has dimension 4 and the disk fiber has dimension 3, the disk bundle has dimension

$$4 + 3 = 7.$$

Therefore

$$\boxed{\mathbb{CP}^3 \text{ bounds a compact 7-manifold.}}$$

6.11 The Diagonal and the Graph of a Map

Let M be a closed F -oriented manifold, and let

$$f : M \rightarrow M$$

be a continuous map.

The *diagonal* is

$$\Delta = \{(p, p) \in M \times M\}.$$

The *graph* of f is

$$\Lambda_f = \{(p, f(p)) \in M \times M\}.$$

A point lies in both Δ and Λ_f precisely when

$$(p, p) = (p, f(p)).$$

That is,

$$f(p) = p.$$

Therefore

$$\Lambda_f \cap \Delta = \{\text{fixed points of } f\}.$$

Thus fixed point theory can be translated into intersection theory. This is the key geometric idea behind the Lefschetz fixed point theorem.

6.12 Poincaré Dual of the Diagonal

Let

$$\{a_i\}$$

be a homogeneous basis of $H^*(M; F)$, and let

$$\{b_i\}$$

be the dual basis with respect to the Poincaré pairing:

$$\langle a_i \smile b_j, [M] \rangle = \delta_{ij}.$$

The Poincaré dual of the diagonal in $M \times M$ is expressed in terms of these bases. A standard formula is

$$PD_{M \times M}(\Delta) = \sum_i (-1)^{|a_i|} p_1^*(a_i) \smile p_2^*(b_i),$$

where

$$p_1, p_2 : M \times M \rightarrow M$$

are the two projections, and $|a_i|$ denotes the degree of a_i .

The sign

$$(-1)^{|a_i|}$$

comes from the Koszul sign convention in graded cohomology.

Conceptually, this formula says that the diagonal class is the cohomological kernel of the identity map, just as

$$\sum_i e_i \otimes e_i^*$$

is the identity tensor in linear algebra.

6.13 Poincaré Dual of the Graph

The graph

$$\Lambda_f = \{(p, f(p))\}$$

is related to the diagonal. Depending on whether one writes the graph as

$$p \mapsto (p, f(p))$$

or as

$$p \mapsto (f(p), p),$$

the pullback f^* appears on the second or first factor.

With the convention

$$\Lambda_f = \{(p, f(p))\},$$

one may write

$$PD(\Lambda_f) = \sum_i (-1)^{|a_i|} p_1^*(a_i) \smile p_2^*(f^*b_i).$$

Equivalently, with the convention

$$\Lambda_f = \{(f(p), p)\},$$

one obtains

$$PD(\Lambda_f) = \sum_i (-1)^{|a_i|} p_1^*(f^*a_i) \smile p_2^*(b_i).$$

Both conventions lead to the same Lefschetz number. The important point is:

the graph class is obtained from the diagonal class by inserting f^* .

6.14 Lefschetz Number

The Lefschetz number of a map

$$f : M \rightarrow M$$

is

$$L(f) = \sum_{j=0}^n (-1)^j \operatorname{Tr} \left(f^* : H^j(M; F) \rightarrow H^j(M; F) \right).$$

This is an alternating sum of traces on cohomology.

It is a homotopy invariant:

$$f \simeq g \implies L(f) = L(g).$$

So $L(f)$ depends only on the induced action of f on cohomology, not on pointwise geometry.

6.15 Lefschetz Number as an Intersection Number

The central result in Problem 11 is

$$L(f) = \Lambda_f \cdot \Delta.$$

The right-hand side is an intersection number in $M \times M$. This statement connects algebra and geometry:

algebraic trace on cohomology = intersection number of graph and diagonal.

If Λ_f and Δ intersect transversely, then

$$\Lambda_f \cdot \Delta$$

is the signed count of fixed points of f . Thus, in favorable cases,

$$L(f) = \sum_{p=f(p)} \text{index}_p(f).$$

Even without transversality, the intersection number remains well-defined algebraically and gives a fixed-point obstruction.

6.16 Lefschetz Fixed Point Theorem

Since fixed points are exactly the intersection points

$$\Lambda_f \cap \Delta,$$

if f has no fixed point, then

$$\Lambda_f \cap \Delta = \emptyset.$$

Therefore

$$\Lambda_f \cdot \Delta = 0.$$

But

$$\Lambda_f \cdot \Delta = L(f).$$

Hence

$$f \text{ has no fixed point} \implies L(f) = 0.$$

Taking the contrapositive gives the Lefschetz fixed point theorem:

$L(f) \neq 0 \implies f \text{ has a fixed point.}$

This is the main conclusion of Problem 11.

6.17 Application to $\mathbb{CP}^2$

The rational cohomology ring of $\mathbb{CP}^2$ is

$$H^*(\mathbb{CP}^2; \mathbb{Q}) \cong \mathbb{Q}[h]/(h^3), \quad \deg h = 2.$$

Thus the only nonzero rational cohomology groups are

$$H^0(\mathbb{CP}^2; \mathbb{Q}) \cong \mathbb{Q},$$

$$H^2(\mathbb{CP}^2; \mathbb{Q}) \cong \mathbb{Q}\langle h \rangle,$$

and

$$H^4(\mathbb{CP}^2; \mathbb{Q}) \cong \mathbb{Q}\langle h^2 \rangle.$$

Any continuous map

$$f : \mathbb{CP}^2 \rightarrow \mathbb{CP}^2$$

induces

$$f^*(h) = d h$$

for some integer d .

Since f^* preserves cup products,

$$f^*(h^2) = f^*(h)^2 = d^2 h^2.$$

Therefore the traces are

$$\mathrm{Tr}(f^*|H^0) = 1,$$

$$\mathrm{Tr}(f^*|H^2) = d,$$

and

$$\mathrm{Tr}(f^*|H^4) = d^2.$$

All these cohomology groups are in even degrees, so

$$L(f) = 1 + d + d^2.$$

But

$$d^2 + d + 1 = \left(d + \frac{1}{2}\right)^2 + \frac{3}{4} > 0.$$

Therefore

$$L(f) \neq 0.$$

By the Lefschetz fixed point theorem, f must have a fixed point. Hence

Every continuous map $f : \mathbb{CP}^2 \rightarrow \mathbb{CP}^2$ has a fixed point.

6.18 Summary of Problems 9–11

The three problems can be summarized as follows:

Problem	Main tool	Main conclusion
9	clutching functions	a nontrivial rank-3 bundle over S^4 has no section
10	Euler characteristic and doubling	$\chi(M) = 0$ for closed odd M , $\chi(M) = \frac{1}{2}\chi(\partial M)$
11	graph-diagonal intersection	$L(f) \neq 0 \Rightarrow f$ has a fixed point

The conceptual chain is:

Problem 9: vector bundle topology $\implies$ sections obstructed by clutching.

Problem 10: manifold topology $\implies$ Euler characteristic obstructs boundaries.

Problem 11: fixed point theory $\implies$ cohomological trace forces intersections.

Thus the common pattern is

algebraic topology converts geometry into computable invariants.

6.19 Figures and Tables for Problems 9–11

The following figures and tables summarize the main mechanisms used in Problems 9–11.

Table 1: Main ingredients in Problem 9. *Source: own material.*

Concept	Role in the construction
Clutching map	A rank-3 bundle over S^4 is built from a map $g : S^3 \rightarrow SO(3)$.
Classification group	Oriented rank-3 bundles over S^4 are classified by $\pi_3(SO(3)) \cong \mathbb{Z}$.
Nonzero section	A nowhere-zero section would reduce the structure group from $SO(3)$ to $SO(2)$.
Obstruction	Since $\pi_3(SO(2)) = 0$, a nonzero class in $\pi_3(SO(3))$ cannot come from $SO(2)$.

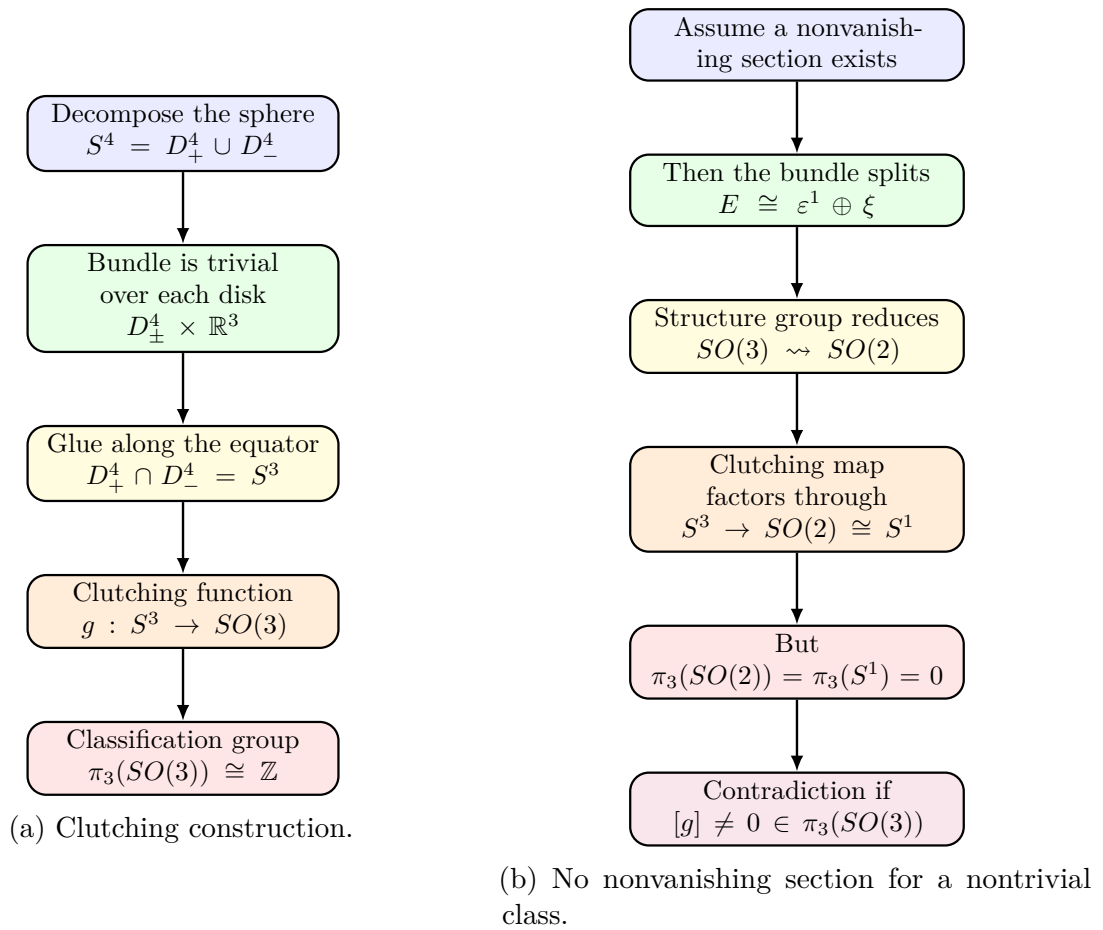


Figure 1: Problem 9 (rank-3 bundles over S^4): clutching construction and the obstruction to a nonvanishing section. *Source: own material.*

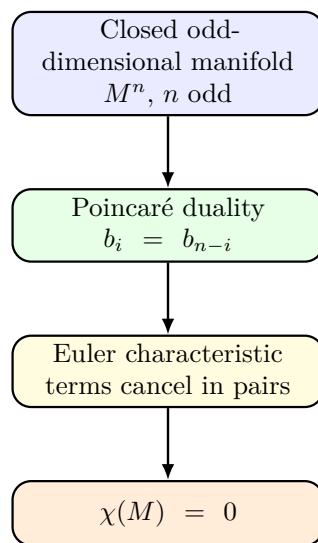


Figure 2: Why closed odd-dimensional manifolds have Euler characteristic zero. *Source: own material.*

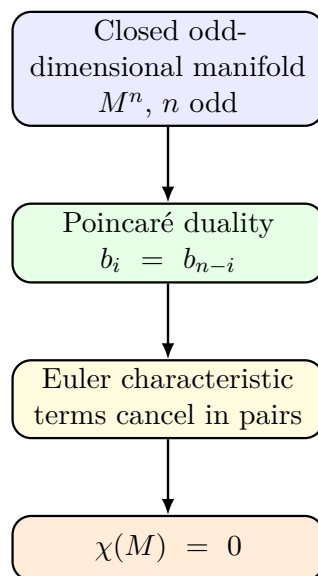


Figure 3: Why closed odd-dimensional manifolds have Euler characteristic zero. *Source: own material.*

Table 2: Main ingredients in Problem 9. *Source: own material.*

Concept	Role in the construction
Clutching map	A rank-3 bundle over S^4 is built from a map $g : S^3 \rightarrow SO(3)$.
Classification group	Oriented rank-3 bundles over S^4 are classified by $\pi_3(SO(3)) \cong \mathbb{Z}$.
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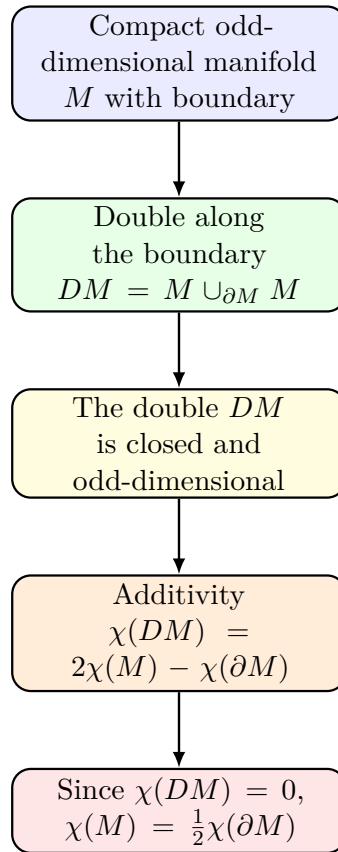


Figure 4: The double construction and the boundary formula for Euler characteristic. *Source: own material.*

Table 3: Boundary obstruction from Euler characteristic in Problem 10. *Source: own material.*

Manifold N	$\chi(N)$	Even?	Conclusion
$\mathbb{RP}^2$	1	no	does not bound a compact 3-manifold
$\mathbb{CP}^2$	3	no	does not bound a compact 5-manifold
$\mathbb{RP}^3$	0	yes	Euler characteristic gives no obstruction; it does bound
$\mathbb{CP}^3$	4	yes	Euler characteristic gives no obstruction; it does bound

Table 4: Examples of compact manifolds bounded by $\mathbb{RP}^3$ and $\mathbb{CP}^3$. *Source: own material.*

Boundary	Bounding manifold	Reason
$\mathbb{RP}^3$	Disk bundle $D(\xi)$ of an oriented 2-plane bundle $\xi \rightarrow S^2$ with $e(\xi) = 2$	$\partial D(\xi) = S(\xi) \cong L(2, 1) \cong \mathbb{RP}^3$
$\mathbb{CP}^3$	Disk bundle $D(\Lambda_+^2(TS^4))$	$\partial D(\Lambda_+^2(TS^4)) \cong S(\Lambda_+^2(TS^4)) \cong \mathbb{CP}^3$

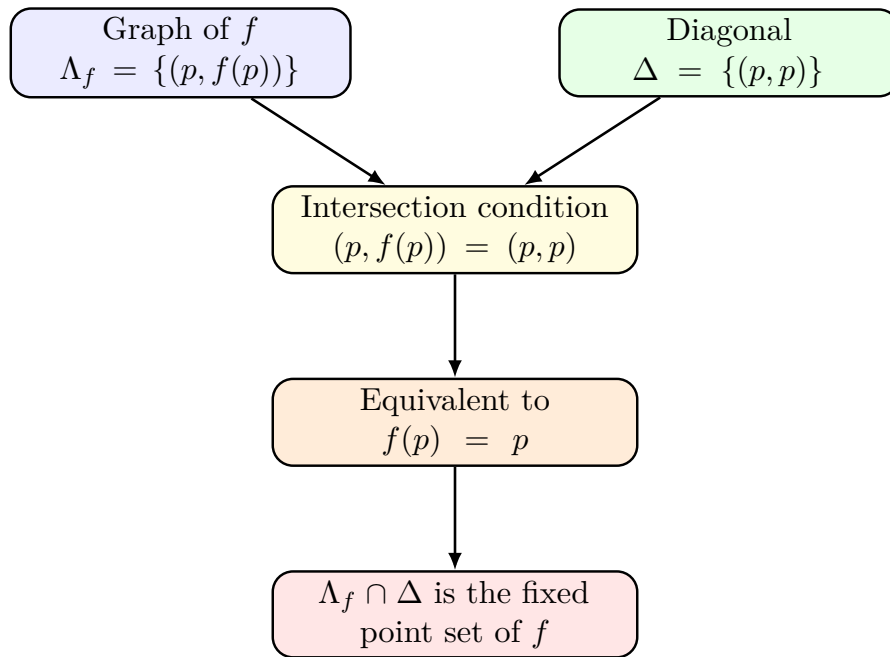


Figure 5: Fixed points as intersections of the graph of f with the diagonal. *Source: own material.*

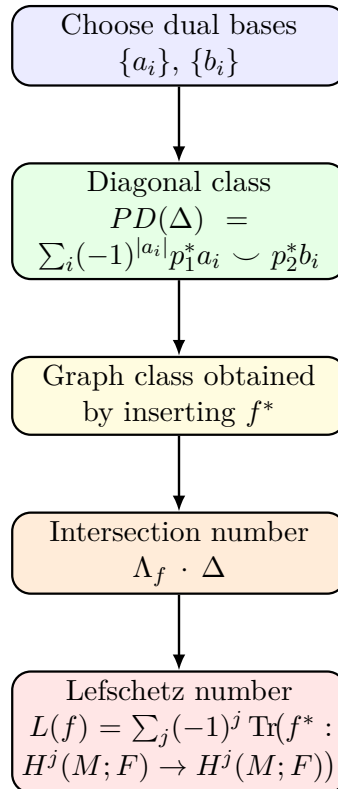
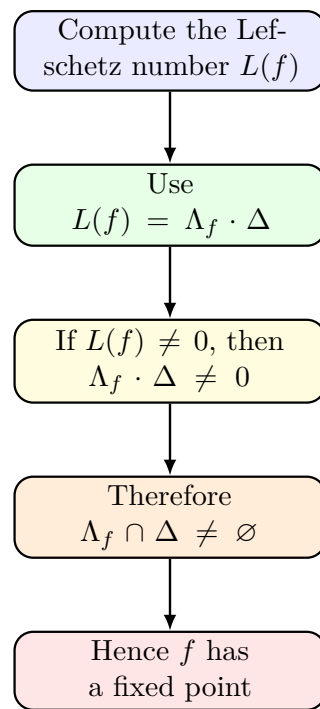
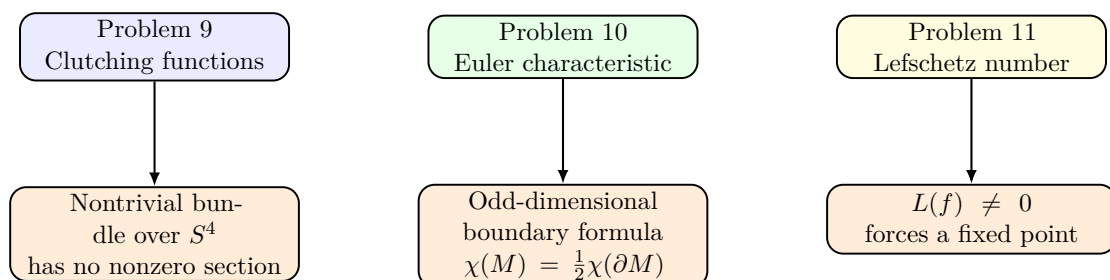


Figure 6: From the diagonal class to the Lefschetz number. *Source: own material.*

Figure 7: Logical structure of the Lefschetz fixed point criterion. *Source: own material.*Table 5: Lefschetz number computation for a map $f : \mathbb{CP}^2 \rightarrow \mathbb{CP}^2$. *Source: own material.*

Degree	Cohomology group	Generator	Trace of f^*
0	$H^0(\mathbb{CP}^2; \mathbb{Q}) \cong \mathbb{Q}$	1	1
2	$H^2(\mathbb{CP}^2; \mathbb{Q}) \cong \mathbb{Q}$	h	d
4	$H^4(\mathbb{CP}^2; \mathbb{Q}) \cong \mathbb{Q}$	h^2	d^2

$$L(f) = 1 + d + d^2 \neq 0.$$

Figure 8: Conceptual summary of Problems 9–11. *Source: own material.*

7 A 3-Dimensional Real Vector Bundle over S^4 with No Nonvanishing Section

7.1 Problem

Construct a real vector bundle

$$E \longrightarrow S^4$$

of rank 3 which has no nonvanishing section.

In other words, we want a 3-dimensional real vector bundle over S^4 such that there is no continuous section

$$s : S^4 \longrightarrow E$$

satisfying

$$s(x) \neq 0 \quad \text{for every } x \in S^4.$$

7.2 Main Idea

The construction uses the *clutching construction*. We decompose the sphere S^4 into two closed 4-balls glued along their common boundary:

$$S^4 = D_+^4 \cup_{S^3} D_-^4.$$

Since each $D_\pm^4$ is contractible, every vector bundle over $D_\pm^4$ is trivial. Therefore, a rank-3 real vector bundle over S^4 can be obtained by taking two trivial bundles

$$D_+^4 \times \mathbb{R}^3 \quad \text{and} \quad D_-^4 \times \mathbb{R}^3,$$

and gluing them along the equator S^3 using a continuous map

$$g : S^3 \longrightarrow GL_3(\mathbb{R}).$$

If we want the bundle to be oriented and equipped with a Euclidean metric, we may take the gluing map to land in $SO(3)$:

$$g : S^3 \longrightarrow SO(3).$$

The resulting bundle is denoted

$$E_g \longrightarrow S^4.$$

The key point is that the homotopy class of the clutching map

$$[g] \in \pi_3(SO(3))$$

determines the isomorphism class of the oriented rank-3 vector bundle over S^4 .

Since

$$\pi_3(SO(3)) \cong \mathbb{Z},$$

there exist nontrivial rank-3 vector bundles over S^4 .

7.3 Step 1: The Clutching Construction

Let

$$S^4 = D_+^4 \cup_{S^3} D_-^4,$$

where

$$\partial D_+^4 = \partial D_-^4 = S^3.$$

Let

$$g : S^3 \longrightarrow SO(3)$$

be a continuous map.

Define

$$E_g = (D_+^4 \times \mathbb{R}^3) \sqcup (D_-^4 \times \mathbb{R}^3) / \sim,$$

where the equivalence relation is given by

$$(q, v)_+ \sim (q, g(q)v)_-, \quad q \in S^3, \quad v \in \mathbb{R}^3.$$

Thus E_g is obtained by gluing two trivial rank-3 bundles along the equator S^3 .

$$E_g = D_+^4 \times \mathbb{R}^3 \cup_g D_-^4 \times \mathbb{R}^3$$

The bundle projection is induced from the natural projections

$$D_\pm^4 \times \mathbb{R}^3 \longrightarrow D_\pm^4.$$

7.4 Step 2: Choosing the Clutching Map

We now choose a specific nontrivial map

$$g : S^3 \longrightarrow SO(3).$$

Identify S^3 with the group of unit quaternions:

$$S^3 = \{q \in \mathbb{H} : |q| = 1\}.$$

Also identify

$$\mathbb{R}^3 \cong \text{Im } \mathbb{H} = \{ai + bj + ck : a, b, c \in \mathbb{R}\}.$$

For $q \in S^3$, define

$$g(q) : \text{Im } \mathbb{H} \longrightarrow \text{Im } \mathbb{H}$$

by

$$g(q)(v) = qvq^{-1}.$$

This is quaternionic conjugation. It preserves the Euclidean norm on $\text{Im } \mathbb{H}$, and it preserves orientation. Hence

$$g(q) \in SO(3).$$

Therefore we obtain a continuous map

$$g : S^3 \longrightarrow SO(3), \quad g(q)(v) = qvq^{-1}.$$

This map is the standard double covering

$$S^3 \cong SU(2) \longrightarrow SO(3).$$

It represents a generator, up to sign, of

$$\pi_3(SO(3)) \cong \mathbb{Z}.$$

Hence

$$[g] \neq 0 \in \pi_3(SO(3)).$$

7.5 Step 3: Definition of the Bundle

Using the above clutching map, define

$$E = (D_+^4 \times \text{Im } \mathbb{H}) \sqcup (D_-^4 \times \text{Im } \mathbb{H}) / \sim,$$

where

$$(q, v)_+ \sim (q, qvq^{-1})_-, \quad q \in S^3, \quad v \in \text{Im } \mathbb{H}.$$

Thus

$$E \longrightarrow S^4$$

is a real vector bundle of rank 3.

Equivalently,

$$E = D_+^4 \times \mathbb{R}^3 \cup_g D_-^4 \times \mathbb{R}^3, \quad g(q)(v) = qvq^{-1}.$$

7.6 Step 4: Why the Bundle is Nontrivial

The isomorphism classes of oriented real rank-3 vector bundles over S^4 are classified by

$$\pi_3(SO(3)).$$

Indeed, the equator in the decomposition

$$S^4 = D_+^4 \cup_{S^3} D_-^4$$

is S^3 , and the transition map along this equator is a map

$$S^3 \rightarrow SO(3).$$

Two clutching maps define isomorphic bundles if and only if they are homotopic. Therefore, since

$$[g] \neq 0 \in \pi_3(SO(3)),$$

the bundle E is nontrivial.

In particular,

$$E \not\cong S^4 \times \mathbb{R}^3.$$

7.7 Step 5: Why a Nonvanishing Section Would Trivialize the Bundle

Suppose, for contradiction, that E admits a nonvanishing section

$$s : S^4 \longrightarrow E.$$

Then for every $x \in S^4$,

$$s(x) \in E_x \quad \text{and} \quad s(x) \neq 0.$$

Since E has a Euclidean metric, we may normalize s . Thus we obtain a unit section

$$\hat{s} : S^4 \longrightarrow E$$

such that

$$\|\hat{s}(x)\| = 1 \quad \text{for every } x \in S^4.$$

This gives a trivial line subbundle

$$\varepsilon^1 \subset E.$$

Since E has rank 3, its orthogonal complement has rank 2. Hence

$$E \cong \varepsilon^1 \oplus F,$$

where $F \rightarrow S^4$ is an oriented real rank-2 vector bundle.

Now oriented real rank-2 vector bundles are classified by their Euler class:

$$e(F) \in H^2(S^4; \mathbb{Z}).$$

But

$$H^2(S^4; \mathbb{Z}) = 0.$$

Therefore

$$e(F) = 0,$$

so F is trivial. Hence

$$F \cong \varepsilon^2.$$

Consequently,

$$E \cong \varepsilon^1 \oplus \varepsilon^2 \cong \varepsilon^3.$$

Thus E would be trivial.

But this contradicts the fact that E was constructed using a nontrivial clutching map

$$[g] \neq 0 \in \pi_3(SO(3)).$$

Therefore E cannot admit a nonvanishing section.

$E \rightarrow S^4 \text{ has no nonvanishing section.}$

7.8 Equivalent Clutching Interpretation of the Obstruction

There is another way to see the same contradiction.

If E had a nowhere-zero section, then the structure group of E would reduce from $SO(3)$ to the subgroup fixing a unit vector. This subgroup is

$$SO(2) \subset SO(3).$$

Therefore the clutching map

$$g : S^3 \longrightarrow SO(3)$$

would be homotopic to a map landing in $SO(2)$:

$$S^3 \longrightarrow SO(2).$$

But

$$SO(2) \cong S^1,$$

and hence

$$\pi_3(SO(2)) = \pi_3(S^1) = 0.$$

So every map

$$S^3 \rightarrow SO(2)$$

is null-homotopic. Hence g would be null-homotopic in $SO(3)$.

This contradicts

$$[g] \neq 0 \in \pi_3(SO(3)).$$

Therefore, again,

$$E$$

has no nonvanishing section.

7.9 Characteristic Class Viewpoint

For an oriented rank-3 real vector bundle

$$E \rightarrow S^4,$$

the Euler class satisfies

$$e(E) \in H^3(S^4; \mathbb{Z}).$$

However,

$$H^3(S^4; \mathbb{Z}) = 0.$$

Therefore

$$e(E) = 0.$$

So the Euler class does not detect the obstruction in this example.

Instead, the obstruction is detected by the first Pontryagin class:

$$p_1(E) \in H^4(S^4; \mathbb{Z}).$$

Since

$$H^4(S^4; \mathbb{Z}) \cong \mathbb{Z},$$

there is a generator

$$u \in H^4(S^4; \mathbb{Z}).$$

For the bundle constructed above, one has

$$p_1(E) = \pm 4u.$$

In particular,

$$p_1(E) \neq 0.$$

If E had a nonvanishing section, then

$$E \cong \varepsilon^1 \oplus F,$$

where F is an oriented rank-2 bundle over S^4 .

But every oriented rank-2 vector bundle over S^4 is trivial, since

$$H^2(S^4; \mathbb{Z}) = 0.$$

Thus $F \cong \varepsilon^2$, and hence

$$E \cong \varepsilon^3.$$

Therefore

$$p_1(E) = 0,$$

which contradicts

$$p_1(E) = \pm 4u \neq 0.$$

Thus the characteristic class viewpoint also proves that E has no nonvanishing section.

7.10 Summary Table

Object	Meaning
$S^4 = D_+^4 \cup_{S^3} D_-^4$	Decomposition of S^4 into two 4-balls glued along the equator.
$D_\pm^4 \times \mathbb{R}^3$	Trivial rank-3 bundles over the two hemispheres.
$g : S^3 \rightarrow SO(3)$	Clutching map used to glue the two trivial bundles.
$g(q)(v) = qvq^{-1}$	Quaternionic conjugation, giving a rotation of $\mathbb{R}^3 \cong \text{Im } \mathbb{H}$.
$[g] \in \pi_3(SO(3))$	Homotopy class classifying the bundle.
$\pi_3(SO(3)) \cong \mathbb{Z}$	There are infinitely many oriented rank-3 bundles over S^4 .
$[g] \neq 0$	The chosen bundle is nontrivial.
Nonvanishing section	Would split E as $\varepsilon^1 \oplus F$.
$H^2(S^4; \mathbb{Z}) = 0$	Forces every oriented rank-2 bundle $F \rightarrow S^4$ to be trivial.
Contradiction	A nonvanishing section would make E trivial, but E is nontrivial.

7.11 Final Answer

The required vector bundle is

$$E = (D_+^4 \times \text{Im } \mathbb{H}) \cup_g (D_-^4 \times \text{Im } \mathbb{H}),$$

where the clutching map

$$g : S^3 \rightarrow SO(3)$$

is defined by

$$g(q)(v) = qvq^{-1}.$$

Since g represents a nonzero element of

$$\pi_3(SO(3)) \cong \mathbb{Z},$$

the bundle E is nontrivial.

If E had a nonvanishing section, then

$$E \cong \varepsilon^1 \oplus F,$$

where F is an oriented rank-2 vector bundle over S^4 . But

$$H^2(S^4; \mathbb{Z}) = 0,$$

so F is trivial, and hence E would be trivial. This contradicts the nontriviality of the clutching map.

Therefore

$$E \rightarrow S^4 \text{ is a real rank-3 vector bundle with no nonvanishing section.}$$

7.12 How to Understand $D_{\pm}^4 \times \mathbb{R}^3$

In the summary table we used the notation

$$D_{\pm}^4 \times \mathbb{R}^3.$$

This means the trivial rank-3 vector bundle over the two hemispheres of S^4 . Let us explain this carefully.

The Sphere S^4

The 4-sphere is defined by

$$S^4 = \{(x_0, x_1, x_2, x_3, x_4) \in \mathbb{R}^5 : x_0^2 + x_1^2 + x_2^2 + x_3^2 + x_4^2 = 1\}.$$

Although S^4 is sitting inside $\mathbb{R}^5$, it is a 4-dimensional manifold.

For comparison:

$$S^1 \subset \mathbb{R}^2, \quad S^2 \subset \mathbb{R}^3, \quad S^3 \subset \mathbb{R}^4, \quad S^4 \subset \mathbb{R}^5.$$

So S^4 is the higher-dimensional analogue of the usual sphere S^2 .

The Two Hemispheres D_+^4 and D_-^4

We split S^4 into two hemispheres:

$$D_+^4 = \{x \in S^4 : x_0 \geq 0\},$$

and

$$D_-^4 = \{x \in S^4 : x_0 \leq 0\}.$$

Each of these spaces is homeomorphic to the closed 4-disk

$$D^4 = \{y \in \mathbb{R}^4 : \|y\| \leq 1\}.$$

Thus

$$D_+^4 \cong D^4, \quad D_-^4 \cong D^4.$$

Their common boundary is the equator:

$$D_+^4 \cap D_-^4 = S^3.$$

Therefore we may write

$$S^4 = D_+^4 \cup_{S^3} D_-^4.$$

This means that S^4 is obtained by gluing two 4-disks along their common boundary S^3 .

$$S^4 = \text{upper 4-disk} \cup_{\text{equator } S^3} \text{lower 4-disk}$$

This is analogous to the familiar decomposition

$$S^2 = D_+^2 \cup_{S^1} D_-^2,$$

where the usual sphere is obtained by gluing two disks along their boundary circle.

The Product $D_{\pm}^4 \times \mathbb{R}^3$

The expression

$$D_{\pm}^4 \times \mathbb{R}^3$$

means the Cartesian product

$$D_{\pm}^4 \times \mathbb{R}^3 = \{(x, v) : x \in D_{\pm}^4, v \in \mathbb{R}^3\}.$$

Here:

$$x \in D_{\pm}^4$$

is a point of the base space, and

$$v \in \mathbb{R}^3$$

is a vector attached to that point.

There is a projection map

$$\pi : D_{\pm}^4 \times \mathbb{R}^3 \longrightarrow D_{\pm}^4$$

defined by

$$\pi(x, v) = x.$$

So the projection forgets the vector v and remembers only the base point x .

Fibers

For a fixed point

$$x \in D_{\pm}^4,$$

the fiber over x is

$$\pi^{-1}(x) = \{(x, v) : v \in \mathbb{R}^3\}.$$

This is naturally identified with

$$\mathbb{R}^3.$$

Thus

$$\pi^{-1}(x) \cong \mathbb{R}^3.$$

So above every point of $D_{\pm}^4$ there is a copy of $\mathbb{R}^3$.

Base point $x \in D_{\pm}^4$ $\longmapsto$ fiber $\mathbb{R}^3$ above x .

Why the Bundle Has Rank 3

A real vector bundle has a real vector space attached to every point of the base.

In our case, the fiber over every point is

$$\mathbb{R}^3.$$

Therefore every fiber has dimension 3. Hence the vector bundle has rank 3.

$$\boxed{\text{rank}(D_{\pm}^4 \times \mathbb{R}^3) = \dim(\mathbb{R}^3) = 3.}$$

The base $D_{\pm}^4$ is 4-dimensional, but the fibers are 3-dimensional.

So one should not confuse:

$$\dim D_{\pm}^4 = 4$$

with

$$\dim \mathbb{R}^3 = 3.$$

The rank of the vector bundle is the dimension of the fiber, not the dimension of the base.

Why the Bundle is Trivial

A vector bundle

$$E \rightarrow B$$

is called trivial if it is globally of the form

$$B \times V,$$

where V is a fixed vector space.

In our case, over each hemisphere we have

$$D_{\pm}^4 \times \mathbb{R}^3.$$

This is already written as a product. Therefore it is a trivial bundle.

$$\boxed{D_{\pm}^4 \times \mathbb{R}^3 \text{ is the trivial rank-3 bundle over } D_{\pm}^4.}$$

There is no twisting inside D_+^4 or inside D_-^4 . The possible twisting appears only when we glue the two trivial bundles along the equator S^3 .

Geometric Interpretation

The expression

$$D_{\pm}^4 \times \mathbb{R}^3$$

should be understood as follows:

$$\boxed{\text{Over every point of the hemisphere } D_{\pm}^4, \text{ place one copy of } \mathbb{R}^3.}$$

So we have:

$$\begin{array}{c}
 \text{base point } x \in D_{\pm}^4 \\
 \downarrow \\
 \text{fiber } \pi^{-1}(x) \cong \mathbb{R}^3.
 \end{array}$$

The total space consists of all pairs

$$(x, v),$$

where x is a point of the hemisphere and v is a vector in the copy of $\mathbb{R}^3$ sitting above x .

Analogy with Lower-Dimensional Cases

It is useful to compare with lower-dimensional examples.

A trivial rank-1 bundle over an interval is

$$[0, 1] \times \mathbb{R}.$$

Above every point of the interval there is a copy of $\mathbb{R}$.

A trivial rank-2 bundle over a disk is

$$D^2 \times \mathbb{R}^2.$$

Above every point of the disk there is a copy of $\mathbb{R}^2$.

In our construction we use

$$D_{\pm}^4 \times \mathbb{R}^3.$$

Above every point of the 4-disk there is a copy of $\mathbb{R}^3$.

Base	Fiber	Trivial bundle
$[0, 1]$	$\mathbb{R}$	$[0, 1] \times \mathbb{R}$
D^2	$\mathbb{R}^2$	$D^2 \times \mathbb{R}^2$
$D_{\pm}^4$	$\mathbb{R}^3$	$D_{\pm}^4 \times \mathbb{R}^3$

Why We Use Trivial Bundles on the Hemispheres

The hemispheres D_+^4 and D_-^4 are contractible. A fundamental fact about vector bundles says:

Every vector bundle over a contractible space is trivial.

Since

$$D_+^4 \simeq \{\text{point}\}, \quad D_-^4 \simeq \{\text{point}\},$$

every rank-3 vector bundle over D_+^4 or D_-^4 is trivial.

Therefore, when constructing a vector bundle over S^4 , the interesting information cannot come from the individual pieces

$$D_+^4 \times \mathbb{R}^3$$

and

$$D_-^4 \times \mathbb{R}^3.$$

The interesting information comes from the gluing along the equator

$$S^3.$$

How the Two Trivial Pieces Are Glued

We begin with two trivial bundles:

$$D_+^4 \times \mathbb{R}^3$$

and

$$D_-^4 \times \mathbb{R}^3.$$

Their boundaries are

$$S^3 \times \mathbb{R}^3$$

on both sides.

To glue them, we choose a map

$$g : S^3 \longrightarrow SO(3).$$

For a point

$$q \in S^3$$

and a vector

$$v \in \mathbb{R}^3,$$

we identify

$$(q, v)_+ \sim (q, g(q)v)_-.$$

Thus the base point q is glued to the same base point q , but the fiber vector v may be rotated by the matrix $g(q)$.

So the gluing rule is

$$\boxed{(q, v)_+ \sim (q, g(q)v)_-}.$$

This is the key mechanism of the clutching construction.

Where the Nontriviality Comes From

Each piece

$$D_+^4 \times \mathbb{R}^3$$

and

$$D_-^4 \times \mathbb{R}^3$$

is trivial.

However, after gluing them together, the resulting bundle over S^4 may be nontrivial.

The nontriviality is encoded by the clutching map

$$g : S^3 \rightarrow SO(3).$$

If g is homotopic to the constant map, then the resulting bundle is trivial.

If g represents a nonzero element of

$$\pi_3(SO(3)),$$

then the resulting bundle is nontrivial.

In our example, we choose

$$g(q)(v) = qvq^{-1},$$

where

$$q \in S^3 \subset \mathbb{H}$$

is a unit quaternion and

$$v \in \text{Im } \mathbb{H} \cong \mathbb{R}^3.$$

This map represents a nonzero element of

$$\pi_3(SO(3)) \cong \mathbb{Z}.$$

Therefore the final bundle is nontrivial.

Short Summary

The notation

$$D_{\pm}^4 \times \mathbb{R}^3$$

means:

the trivial rank-3 vector bundle over the upper or lower hemisphere of S^4 .

More explicitly,

$$D_{\pm}^4 \times \mathbb{R}^3 = \{(x, v) : x \in D_{\pm}^4, v \in \mathbb{R}^3\}.$$

The projection is

$$\pi(x, v) = x.$$

The fiber over x is

$$\pi^{-1}(x) = \{x\} \times \mathbb{R}^3 \cong \mathbb{R}^3.$$

Thus above every point of the hemisphere $D_{\pm}^4$ we place one copy of $\mathbb{R}^3$.

The bundle is trivial over each hemisphere, but the final bundle over S^4 can be nontrivial because the two trivial pieces are glued along the equator S^3 using a nontrivial map

$$g : S^3 \rightarrow SO(3).$$

8 Rank 1, Rank 2, and Rank 3 Real Vector Bundles

8.1 Definition of a Real Vector Bundle

A real vector bundle is a continuous map

$$\pi : E \longrightarrow B$$

such that for every point $x \in B$, the fiber

$$E_x = \pi^{-1}(x)$$

is a real vector space, and locally the bundle looks like a product.

More precisely, for every point $x \in B$, there exists an open neighborhood

$$U \subset B$$

such that

$$\pi^{-1}(U) \cong U \times \mathbb{R}^r.$$

The number r is called the *rank* of the vector bundle.

Thus

$$\boxed{\text{rank}(E) = \dim(E_x).}$$

The rank is the dimension of the fiber, not the dimension of the base.

8.2 Basic Interpretation

A vector bundle should be understood as follows:

$$\boxed{\text{Over every point } x \in B, \text{ there is a vector space } E_x.}$$

So the base space B tells us where the fibers are placed, and the fibers tell us what vector spaces sit above the points of B .

For a rank r real vector bundle,

$$E_x \cong \mathbb{R}^r$$

for every $x \in B$.

Therefore:

Rank	Fiber	Name
1	$\mathbb{R}$	real line bundle
2	$\mathbb{R}^2$	real plane bundle
3	$\mathbb{R}^3$	rank-3 real vector bundle

8.3 The Trivial Rank- r Bundle

The simplest vector bundle over a space B is the product bundle

$$B \times \mathbb{R}^r \longrightarrow B.$$

The projection is

$$\pi : B \times \mathbb{R}^r \longrightarrow B, \quad \pi(x, v) = x.$$

The fiber over $x \in B$ is

$$\pi^{-1}(x) = \{x\} \times \mathbb{R}^r.$$

Hence

$$\pi^{-1}(x) \cong \mathbb{R}^r.$$

This bundle is called the *trivial rank- r real vector bundle* over B . It is often denoted by

$$\varepsilon_B^r.$$

Thus

$$\varepsilon_B^r = B \times \mathbb{R}^r.$$

For example,

$$\varepsilon_B^1 = B \times \mathbb{R},$$

$$\varepsilon_B^2 = B \times \mathbb{R}^2,$$

and

$$\varepsilon_B^3 = B \times \mathbb{R}^3.$$

8.4 Rank 1 Vector Bundles: Line Bundles

A rank 1 real vector bundle is called a *real line bundle*. Its fibers are copies of $\mathbb{R}$.

Thus, if

$$\pi : L \rightarrow B$$

is a real line bundle, then for every $x \in B$,

$$L_x = \pi^{-1}(x) \cong \mathbb{R}.$$

So above every point of B , there is one real line.

Example: The Trivial Line Bundle over S^1

The trivial line bundle over the circle is

$$S^1 \times \mathbb{R} \longrightarrow S^1.$$

The projection is

$$\pi(x, t) = x.$$

The fiber over $x \in S^1$ is

$$\pi^{-1}(x) = \{x\} \times \mathbb{R} \cong \mathbb{R}.$$

Geometrically, the total space looks like a cylinder.

This bundle has a nonvanishing section

$$s : S^1 \rightarrow S^1 \times \mathbb{R}$$

defined by

$$s(x) = (x, 1).$$

Indeed,

$$s(x) \neq 0$$

in every fiber.

Therefore

$$S^1 \times \mathbb{R}$$

is a trivial rank 1 bundle.

Example: The Möbius Line Bundle

The Möbius band is also a rank 1 real vector bundle over S^1 .

Its projection is

$$\pi : M \rightarrow S^1,$$

where M denotes the Möbius band.

For every point $x \in S^1$, the fiber is a line:

$$M_x \cong \mathbb{R}.$$

Locally, the Möbius band looks like a product:

$$U \times \mathbb{R}.$$

However, globally it is twisted. It is not isomorphic to

$$S^1 \times \mathbb{R}.$$

Thus

The Möbius band is a nontrivial real line bundle over S^1 .

The geometric reason is that when one goes once around the circle, the fiber direction is reversed.

In clutching language, the transition function takes values in

$$O(1) = \{\pm 1\}.$$

The trivial line bundle corresponds to gluing by $+1$, while the Möbius bundle corresponds to gluing by -1 .

8.5 Rank 2 Vector Bundles: Plane Bundles

A rank 2 real vector bundle is often called a *real plane bundle*. Its fibers are copies of $\mathbb{R}^2$. Thus, if

$$\pi : E \rightarrow B$$

is a rank 2 real vector bundle, then

$$E_x \cong \mathbb{R}^2$$

for every $x \in B$.

So above every point of the base space, there is a plane.

Example: The Trivial Rank 2 Bundle

The trivial rank 2 bundle over B is

$$B \times \mathbb{R}^2 \longrightarrow B.$$

The projection is

$$\pi(x, v) = x.$$

The fiber over $x \in B$ is

$$\pi^{-1}(x) = \{x\} \times \mathbb{R}^2 \cong \mathbb{R}^2.$$

This bundle has two globally defined sections

$$s_1(x) = (x, (1, 0)),$$

and

$$s_2(x) = (x, (0, 1)).$$

For every $x \in B$, the vectors

$$s_1(x), s_2(x)$$

form a basis of the fiber

$$E_x \cong \mathbb{R}^2.$$

Therefore

$$B \times \mathbb{R}^2$$

is a trivial rank 2 bundle.

Example: The Tangent Bundle of S^2

The tangent bundle of the 2-sphere is

$$TS^2 \longrightarrow S^2.$$

For each point $p \in S^2$, the fiber is the tangent plane at p :

$$(TS^2)_p = T_p S^2.$$

Since S^2 is 2-dimensional,

$$T_p S^2 \cong \mathbb{R}^2.$$

Therefore

$$TS^2$$

is a rank 2 real vector bundle over S^2 .

However, TS^2 is not trivial. If it were trivial, then there would exist two everywhere linearly independent tangent vector fields on S^2 . In particular, there would exist one nowhere-zero tangent vector field on S^2 .

But the hairy ball theorem says that every continuous tangent vector field on S^2 must vanish somewhere.

Hence

$$TS^2 \rightarrow S^2 \text{ is a nontrivial rank 2 real vector bundle.}$$

Example: The Tangent Bundle of S^1

The tangent bundle of the circle is

$$TS^1 \longrightarrow S^1.$$

Since S^1 is 1-dimensional, each tangent space is

$$T_p S^1 \cong \mathbb{R}.$$

Therefore

$$TS^1$$

is a rank 1 bundle, not a rank 2 bundle.

In fact,

$$TS^1 \cong S^1 \times \mathbb{R}.$$

Thus the tangent bundle of the circle is trivial.

8.6 Rank 3 Vector Bundles

A rank 3 real vector bundle has fibers isomorphic to

$$\mathbb{R}^3.$$

Thus, if

$$\pi : E \rightarrow B$$

is a rank 3 real vector bundle, then for every point $x \in B$,

$$E_x \cong \mathbb{R}^3.$$

So above every point of B , there is a 3-dimensional real vector space.

Example: The Trivial Rank 3 Bundle

The trivial rank 3 bundle over B is

$$B \times \mathbb{R}^3 \longrightarrow B.$$

The projection is

$$\pi(x, v) = x.$$

The fiber over $x \in B$ is

$$\pi^{-1}(x) = \{x\} \times \mathbb{R}^3 \cong \mathbb{R}^3.$$

This bundle has three globally defined sections:

$$s_1(x) = (x, (1, 0, 0)),$$

$$s_2(x) = (x, (0, 1, 0)),$$

and

$$s_3(x) = (x, (0, 0, 1)).$$

For every $x \in B$, these three vectors form a basis of the fiber:

$$s_1(x), s_2(x), s_3(x)$$

is a basis of

$$E_x \cong \mathbb{R}^3.$$

Therefore

$$B \times \mathbb{R}^3$$

is a trivial rank 3 vector bundle.

Example: The Tangent Bundle of S^3

The tangent bundle of S^3 is

$$TS^3 \longrightarrow S^3.$$

Since S^3 is 3-dimensional, each tangent space satisfies

$$T_p S^3 \cong \mathbb{R}^3.$$

Therefore

$$TS^3$$

is a rank 3 real vector bundle.

A special fact is that S^3 is parallelizable. This means that its tangent bundle is trivial:

$$TS^3 \cong S^3 \times \mathbb{R}^3.$$

Hence

$TS^3 \rightarrow S^3$ is a trivial rank 3 real vector bundle.
--

Example: A Nontrivial Rank 3 Bundle over S^4

In the previous construction, we considered a rank 3 real vector bundle

$$E \longrightarrow S^4.$$

Here the base is S^4 , but the fibers are copies of $\mathbb{R}^3$:

$$E_x \cong \mathbb{R}^3.$$

We construct this bundle by decomposing

$$S^4 = D_+^4 \cup_{S^3} D_-^4.$$

Over each hemisphere, the bundle is trivial:

$$D_+^4 \times \mathbb{R}^3$$

and

$$D_-^4 \times \mathbb{R}^3.$$

Then we glue these two trivial bundles along the equator S^3 using a clutching map

$$g : S^3 \rightarrow SO(3).$$

Thus

$$E = D_+^4 \times \mathbb{R}^3 \cup_g D_-^4 \times \mathbb{R}^3.$$

In our example, we choose

$$g(q)(v) = qvq^{-1},$$

where

$$q \in S^3 \subset \mathbb{H}$$

is a unit quaternion and

$$v \in \text{Im } \mathbb{H} \cong \mathbb{R}^3.$$

This clutching map represents a nonzero element of

$$\pi_3(SO(3)) \cong \mathbb{Z}.$$

Therefore the resulting rank 3 vector bundle over S^4 is nontrivial.

Moreover, this bundle has no nonvanishing section.

8.7 Sections of Vector Bundles

Let

$$\pi : E \rightarrow B$$

be a vector bundle.

A section is a continuous map

$$s : B \rightarrow E$$

such that

$$\pi(s(x)) = x$$

for every $x \in B$.

Equivalently, a section chooses one vector from each fiber:

$$s(x) \in E_x.$$

A section is called nonvanishing if

$$s(x) \neq 0$$

for every $x \in B$.

Example: Sections of a Trivial Bundle

For the trivial bundle

$$E = B \times \mathbb{R}^r,$$

a section has the form

$$s(x) = (x, f(x)),$$

where

$$f : B \rightarrow \mathbb{R}^r$$

is a continuous map.

For example, in the trivial rank 3 bundle

$$B \times \mathbb{R}^3,$$

the section

$$s(x) = (x, (1, 0, 0))$$

is nonvanishing.

Example: Sections of a Tangent Bundle

For a tangent bundle

$$TM \rightarrow M,$$

a section is exactly a vector field on M .

Indeed, a section

$$s : M \rightarrow TM$$

assigns to every point $p \in M$ a tangent vector

$$s(p) \in T_p M.$$

Thus

$$\boxed{\text{Sections of } TM = \text{vector fields on } M.}$$

A nonvanishing section of TM is a vector field which is never zero.

8.8 Local Triviality

A vector bundle does not need to be globally a product. However, it must be locally a product.

This means that for each point $x \in B$, there exists an open neighborhood $U \subset B$ such that

$$\pi^{-1}(U) \cong U \times \mathbb{R}^r.$$

Therefore every vector bundle looks locally like a trivial bundle.

The difference between a trivial and nontrivial bundle is global.

For example, the Möbius band is locally a product:

$$U \times \mathbb{R},$$

but globally it is twisted and is not isomorphic to

$$S^1 \times \mathbb{R}.$$

Thus

$$\boxed{\text{Vector bundles are locally trivial but may be globally nontrivial.}}$$

8.9 Transition Functions

Let

$$\pi : E \rightarrow B$$

be a rank r real vector bundle.

Choose open sets

$$U_i \subset B$$

such that

$$\pi^{-1}(U_i) \cong U_i \times \mathbb{R}^r.$$

On an overlap

$$U_i \cap U_j,$$

the two trivializations may differ by an invertible linear map.

Therefore we obtain transition functions

$$g_{ij} : U_i \cap U_j \rightarrow GL_r(\mathbb{R}).$$

These functions describe how the local product pieces are glued together.

If the vector bundle is equipped with a Euclidean metric, the structure group may be reduced to

$$O(r).$$

If the bundle is additionally oriented, the structure group may be reduced to

$$SO(r).$$

For rank 3, the oriented transition functions take values in

$$SO(3).$$

Thus a rank 3 oriented real vector bundle is locally glued by maps

$$g_{ij} : U_i \cap U_j \rightarrow SO(3).$$

8.10 Triviality and Global Frames

A rank r vector bundle

$$E \rightarrow B$$

is trivial if and only if it admits r everywhere linearly independent global sections.

That means there exist sections

$$s_1, \dots, s_r : B \rightarrow E$$

such that, for every $x \in B$,

$$s_1(x), \dots, s_r(x)$$

form a basis of the fiber

$$E_x.$$

Such a collection is called a global frame.

Therefore:

Rank 1 bundle is trivial $\iff$ it has one nonvanishing section.

Rank 2 bundle is trivial $\iff$ it has two everywhere independent sections.

Rank 3 bundle is trivial $\iff$ it has three everywhere independent sections.

For a rank 3 bundle, one nonvanishing section does not generally trivialize the whole bundle. It only splits off a trivial line bundle:

$$E \cong \varepsilon^1 \oplus F,$$

where F is a rank 2 vector bundle.

However, in the special case of our bundle over S^4 , if such a section existed, then F would be an oriented rank 2 vector bundle over S^4 . Since

$$H^2(S^4; \mathbb{Z}) = 0,$$

the bundle F would be trivial. Hence E would be trivial.

This contradicts the nontrivial clutching construction. Therefore our rank 3 bundle over S^4 has no nonvanishing section.

8.11 Comparison Table

Rank	Name	Fiber	Typical Example
1	Line bundle	$\mathbb{R}$	Möbius band over S^1
2	Plane bundle	$\mathbb{R}^2$	$TS^2 \rightarrow S^2$
3	Rank-3 bundle	$\mathbb{R}^3$	Clutched bundle $E \rightarrow S^4$

8.12 Examples Table

Bundle	Base	Fiber	Trivial?
$S^1 \times \mathbb{R}$	S^1	$\mathbb{R}$	Yes
Möbius band	S^1	$\mathbb{R}$	No
$S^2 \times \mathbb{R}^2$	S^2	$\mathbb{R}^2$	Yes
TS^2	S^2	$\mathbb{R}^2$	No
$S^3 \times \mathbb{R}^3$	S^3	$\mathbb{R}^3$	Yes
TS^3	S^3	$\mathbb{R}^3$	Yes
$E \rightarrow S^4$	S^4	$\mathbb{R}^3$	No

8.13 Final Summary

A real vector bundle is a family of vector spaces varying continuously over a base space.

Base space B + fiber $\mathbb{R}^r$ = rank r real vector bundle.
--

Rank 1 bundles have fibers $\mathbb{R}$, so they are line bundles.

Rank 2 bundles have fibers $\mathbb{R}^2$, so they are plane bundles.

Rank 3 bundles have fibers $\mathbb{R}^3$.

A bundle is trivial if all fibers fit together globally as a product:

$$B \times \mathbb{R}^r.$$

A bundle may be locally trivial but globally nontrivial. The Möbius band is the basic rank 1 example, TS^2 is a basic rank 2 example, and the clutched bundle

$$E = D_+^4 \times \mathbb{R}^3 \cup_g D_-^4 \times \mathbb{R}^3$$

with

$$g(q)(v) = qvq^{-1}$$

is a rank 3 example over S^4 with no nonvanishing section.

8.14 The Case $B = S^2$: Trivial Plane Bundle versus Tangent Bundle

Let

$$B = S^2.$$

A natural first idea is to take the product bundle

$$B \times \mathbb{R}^2 \longrightarrow B.$$

Since $B = S^2$, this becomes

$$S^2 \times \mathbb{R}^2 \longrightarrow S^2.$$

The projection is

$$\pi : S^2 \times \mathbb{R}^2 \longrightarrow S^2, \quad \pi(p, v) = p.$$

For each point $p \in S^2$, the fiber is

$$\pi^{-1}(p) = \{p\} \times \mathbb{R}^2.$$

Thus

$$\pi^{-1}(p) \cong \mathbb{R}^2.$$

Therefore

$$S^2 \times \mathbb{R}^2 \rightarrow S^2$$

is a rank 2 real vector bundle.

However, it is important to distinguish this trivial bundle from the tangent bundle of the sphere.

The Trivial Plane Bundle over S^2

The bundle

$$S^2 \times \mathbb{R}^2 \longrightarrow S^2$$

means that above every point $p \in S^2$, we attach the same fixed model vector space

$$\mathbb{R}^2.$$

So we should think of it as

At every point $p \in S^2$, place one fixed copy of $\mathbb{R}^2$.

The total space consists of pairs

$$(p, v),$$

where

$$p \in S^2, \quad v \in \mathbb{R}^2.$$

This bundle is globally a product, so it is trivial:

$S^2 \times \mathbb{R}^2$ is the trivial rank 2 bundle over S^2 .

It has two globally defined linearly independent sections:

$$s_1 : S^2 \rightarrow S^2 \times \mathbb{R}^2, \quad s_1(p) = (p, (1, 0)),$$

and

$$s_2 : S^2 \rightarrow S^2 \times \mathbb{R}^2, \quad s_2(p) = (p, (0, 1)).$$

For every point $p \in S^2$, the two vectors

$$s_1(p), s_2(p)$$

form a basis of the fiber

$$\{p\} \times \mathbb{R}^2.$$

Hence

$$s_1, s_2$$

form a global frame. This is exactly what it means for a rank 2 bundle to be trivial.

The Tangent Bundle of S^2

The tangent bundle of the sphere is written

$$TS^2 \longrightarrow S^2.$$

Its fiber over a point $p \in S^2$ is the tangent plane at p :

$$(TS^2)_p = T_p S^2.$$

Since S^2 is a 2-dimensional manifold, each tangent plane is a 2-dimensional real vector space:

$$T_p S^2 \cong \mathbb{R}^2.$$

Therefore

$$TS^2$$

is also a rank 2 real vector bundle.

But now the fiber over p is not just a fixed abstract copy of $\mathbb{R}^2$. It is the actual tangent plane to the sphere at p .

A useful concrete description is:

$$TS^2 = \{(p, v) \in S^2 \times \mathbb{R}^3 : p \cdot v = 0\}.$$

Here $S^2 \subset \mathbb{R}^3$, so a point

$$p \in S^2$$

is a unit vector in $\mathbb{R}^3$, and

$$v \in \mathbb{R}^3.$$

The condition

$$p \cdot v = 0$$

means that v is perpendicular to p . Since the radius vector p is normal to the sphere at p , the vectors perpendicular to p form the tangent plane:

$$T_p S^2 = \{v \in \mathbb{R}^3 : p \cdot v = 0\}.$$

The projection is

$$\pi : TS^2 \longrightarrow S^2, \quad \pi(p, v) = p.$$

The fiber over p is

$$\pi^{-1}(p) = \{(p, v) : v \in \mathbb{R}^3, p \cdot v = 0\}.$$

Thus

$$\pi^{-1}(p) \cong T_p S^2.$$

So we have

$$(TS^2)_p = T_p S^2 = \{v \in \mathbb{R}^3 : p \cdot v = 0\}.$$

Why TS^2 is Not the Same as $S^2 \times \mathbb{R}^2$

Both bundles

$$S^2 \times \mathbb{R}^2 \rightarrow S^2$$

and

$$TS^2 \rightarrow S^2$$

have fibers isomorphic to $\mathbb{R}^2$.

However, they are not the same vector bundle.

The trivial bundle

$$S^2 \times \mathbb{R}^2$$

has one fixed copy of $\mathbb{R}^2$ over every point.

The tangent bundle

$$TS^2$$

has the tangent plane

$$T_p S^2$$

over each point p . These tangent planes rotate as p moves around the sphere.

Locally, TS^2 looks like a product:

$$\pi^{-1}(U) \cong U \times \mathbb{R}^2$$

for sufficiently small open sets

$$U \subset S^2.$$

But globally,

$$TS^2$$

is twisted.

In fact,

$$TS^2 \not\cong S^2 \times \mathbb{R}^2.$$

One reason is the hairy ball theorem:

Every continuous tangent vector field on S^2 vanishes somewhere.

If

$$TS^2 \cong S^2 \times \mathbb{R}^2,$$

then TS^2 would have a global frame, and in particular it would have a nowhere-zero tangent vector field.

But the hairy ball theorem says that this is impossible.

Therefore

$$TS^2$$

is nontrivial.

Comparison

The two bundles should be compared as follows:

Bundle	Base	Fiber over p	Trivial?
$S^2 \times \mathbb{R}^2 \rightarrow S^2$	S^2	$\{p\} \times \mathbb{R}^2 \cong \mathbb{R}^2$	Yes
$TS^2 \rightarrow S^2$	S^2	$T_p S^2 \cong \mathbb{R}^2$	No

Thus we have the important distinction:

Every tangent plane $T_p S^2$ is isomorphic to $\mathbb{R}^2$,

but

the whole tangent bundle TS^2 is not globally $S^2 \times \mathbb{R}^2$.

Best Summary

The trivial rank 2 bundle over S^2 is

$$S^2 \times \mathbb{R}^2 \rightarrow S^2.$$

It means:

same fixed plane $\mathbb{R}^2$ over every point of S^2 .

The tangent bundle of S^2 is

$$TS^2 \rightarrow S^2.$$

It means:

the actual tangent plane $T_p S^2$ over each point $p \in S^2$.

Although

$$T_p S^2 \cong \mathbb{R}^2$$

for each individual point p , the bundle as a whole is twisted:

$$TS^2 \not\cong S^2 \times \mathbb{R}^2.$$

9 Euler Characteristic of Closed Odd-Dimensional Manifolds

9.1 Basic definitions

Definition 9.1 (Manifold). An n -dimensional manifold is a topological space M such that every point $p \in M$ has a neighbourhood homeomorphic to an open subset of $\mathbb{R}^n$. In other words, locally M looks like $\mathbb{R}^n$.

Example 9.2. The circle S^1 is a 1-dimensional manifold, because near every point it looks like an open interval in $\mathbb{R}$.

Example 9.3. The sphere S^2 is a 2-dimensional manifold, because near every point it looks like an open disk in $\mathbb{R}^2$.

Example 9.4. The 3-sphere S^3 is a 3-dimensional manifold. Although it is harder to visualize, locally every point has a neighbourhood homeomorphic to an open ball in $\mathbb{R}^3$.

Definition 9.5 (Closed manifold). A manifold M is called *closed* if it is compact and has no boundary.

Example 9.6. The circle S^1 is closed. It is compact and has no boundary.

Example 9.7. The interval $[0, 1]$ is compact, but it is not closed as a manifold without boundary, because it has boundary points 0 and 1.

Example 9.8. The sphere S^2 is closed. The disk D^2 is not closed as a manifold without boundary, because its boundary is S^1 .

9.2 Betti numbers

Let M be a topological space. Its homology groups $H_i(M; F)$ measure i -dimensional holes in M , where F is a field.

Definition 9.9 (Betti number). The i -th Betti number of M over a field F is

$$b_i(M) = \dim_F H_i(M; F).$$

Intuitively:

$b_0(M)$	number of connected components,
$b_1(M)$	number of independent loops,
$b_2(M)$	number of independent enclosed voids or surfaces,
$b_i(M)$	number of independent i -dimensional holes.

Example 9.10. For the circle S^1 we have

$$b_0(S^1) = 1, \quad b_1(S^1) = 1.$$

There is one connected component and one independent loop.

Example 9.11. For the sphere S^2 we have

$$b_0(S^2) = 1, \quad b_1(S^2) = 0, \quad b_2(S^2) = 1.$$

There is one connected component, no tunnel, and one 2-dimensional spherical surface class.

9.3 Euler characteristic

Definition 9.12 (Euler characteristic). Let M be a space with finite-dimensional homology groups. The Euler characteristic of M is defined by

$$\chi(M) = \sum_{i=0}^n (-1)^i b_i(M).$$

Thus the Euler characteristic is the alternating sum of Betti numbers:

$$\chi(M) = b_0(M) - b_1(M) + b_2(M) - b_3(M) + \cdots + (-1)^n b_n(M).$$

9.4 Step 1: Closed odd-dimensional manifolds

Let M be a closed n -dimensional manifold, where n is odd. We compute the Euler characteristic using Betti numbers over a field F :

$$\chi(M) = \sum_{i=0}^n (-1)^i b_i(M).$$

The key fact is Poincaré duality.

Theorem 9.1 (Poincaré duality). If M is a closed n -dimensional manifold, then the Betti numbers satisfy

$$b_i(M) = b_{n-i}(M).$$

If M is orientable, this holds over any field F . For a general closed manifold, one may work over $\mathbb{F}_2$.

Now suppose that n is odd. Pair the terms i and $n - i$ in the Euler characteristic:

$$(-1)^i b_i(M) + (-1)^{n-i} b_{n-i}(M).$$

By Poincaré duality,

$$b_{n-i}(M) = b_i(M).$$

Therefore the paired terms become

$$(-1)^i b_i(M) + (-1)^{n-i} b_i(M).$$

Since n is odd, the signs $(-1)^i$ and $(-1)^{n-i}$ are opposite. Hence

$$(-1)^i + (-1)^{n-i} = 0.$$

So each pair cancels:

$$(-1)^i b_i(M) + (-1)^{n-i} b_{n-i}(M) = 0.$$

Therefore,

$$\boxed{\chi(M) = 0.}$$

So every closed odd-dimensional manifold has Euler characteristic zero.

9.5 Example 1: The circle S^1

The circle S^1 is a closed 1-dimensional manifold. Its Betti numbers are

$$b_0(S^1) = 1, \quad b_1(S^1) = 1.$$

Therefore

$$\chi(S^1) = b_0(S^1) - b_1(S^1) = 1 - 1 = 0.$$

Thus

$$\boxed{\chi(S^1) = 0.}$$

This agrees with the theorem, because S^1 is closed and odd-dimensional.

9.6 Example 2: The 3-sphere S^3

The 3-sphere S^3 is a closed 3-dimensional manifold. Its Betti numbers are

$$b_0(S^3) = 1, \quad b_1(S^3) = 0, \quad b_2(S^3) = 0, \quad b_3(S^3) = 1.$$

Hence

$$\chi(S^3) = b_0(S^3) - b_1(S^3) + b_2(S^3) - b_3(S^3).$$

Substituting the Betti numbers gives

$$\chi(S^3) = 1 - 0 + 0 - 1 = 0.$$

Thus

$$\boxed{\chi(S^3) = 0.}$$

Again, this matches the general theorem.

9.7 Example 3: The 3-torus T^3

The 3-torus is

$$T^3 = S^1 \times S^1 \times S^1.$$

It is a closed 3-dimensional manifold. Its Betti numbers are

$$b_0(T^3) = 1, \quad b_1(T^3) = 3, \quad b_2(T^3) = 3, \quad b_3(T^3) = 1.$$

Therefore

$$\chi(T^3) = b_0(T^3) - b_1(T^3) + b_2(T^3) - b_3(T^3).$$

So

$$\chi(T^3) = 1 - 3 + 3 - 1 = 0.$$

Thus

$$\boxed{\chi(T^3) = 0.}$$

This is another example of a closed odd-dimensional manifold with Euler characteristic zero.

9.8 Comparison with even-dimensional manifolds

The cancellation argument depends on n being odd. In even dimension, the signs of paired terms do not cancel in the same way.

For example, the sphere S^2 has Betti numbers

$$b_0(S^2) = 1, \quad b_1(S^2) = 0, \quad b_2(S^2) = 1.$$

Hence

$$\chi(S^2) = 1 - 0 + 1 = 2.$$

So

$$\boxed{\chi(S^2) = 2.}$$

Thus closed even-dimensional manifolds do not necessarily have Euler characteristic zero.

9.9 Summary

For a closed n -dimensional manifold M , Poincaré duality gives

$$b_i(M) = b_{n-i}(M).$$

If n is odd, then the Euler characteristic terms cancel in pairs:

$$(-1)^i b_i(M) + (-1)^{n-i} b_{n-i}(M) = 0.$$

Therefore,

If M is a closed odd-dimensional manifold, then $\chi(M) = 0$.

10 Betti Numbers and Euler Characteristic: Definitions and Examples

10.1 Definition of Betti numbers

Let X be a topological space and let F be a field, for example $F = \mathbb{Q}$, $\mathbb{R}$, or $\mathbb{F}_2$.

Definition 10.1 (Betti number). The i -th Betti number of X over the field F is defined as

$$b_i(X) = \dim_F H_i(X; F),$$

where $H_i(X; F)$ is the i -th homology group of X with coefficients in F .

Intuitively, Betti numbers count independent holes of different dimensions:

$b_0(X)$	number of connected components,
$b_1(X)$	number of independent loops,
$b_2(X)$	number of independent enclosed surfaces or voids,
$b_3(X)$	number of independent 3-dimensional holes.

10.2 Definition of Euler characteristic

Definition 10.2 (Euler characteristic). Let X be a space whose homology groups are finite-dimensional over F . The Euler characteristic of X is

$$\chi(X) = \sum_{i \geq 0} (-1)^i b_i(X).$$

Equivalently, if X has dimension n , then

$$\chi(X) = b_0(X) - b_1(X) + b_2(X) - b_3(X) + \cdots + (-1)^n b_n(X).$$

Thus the Euler characteristic is an alternating sum of Betti numbers.

10.3 Example 1: The circle S^1

The circle S^1 is connected, so

$$b_0(S^1) = 1.$$

It has one independent loop, namely the loop going once around the circle, so

$$b_1(S^1) = 1.$$

There are no higher-dimensional holes, so

$$b_i(S^1) = 0 \quad \text{for } i \geq 2.$$

Therefore the only nonzero Betti numbers are

$$b_0(S^1) = 1, \quad b_1(S^1) = 1.$$

Now compute the Euler characteristic:

$$\chi(S^1) = b_0(S^1) - b_1(S^1).$$

Substituting the Betti numbers gives

$$\chi(S^1) = 1 - 1 = 0.$$

Hence

$$\boxed{\chi(S^1) = 0.}$$

So the circle has Euler characteristic zero.

10.4 Example 2: The sphere S^2

The sphere S^2 is connected, hence

$$b_0(S^2) = 1.$$

Every loop on S^2 can be contracted to a point. Therefore there are no independent 1-dimensional loops:

$$b_1(S^2) = 0.$$

However, the whole sphere represents one fundamental 2-dimensional homology class. Hence

$$b_2(S^2) = 1.$$

There are no higher-dimensional homology groups, so

$$b_i(S^2) = 0 \quad \text{for } i \geq 3.$$

Thus the Betti numbers are

$$b_0(S^2) = 1, \quad b_1(S^2) = 0, \quad b_2(S^2) = 1.$$

Now compute:

$$\chi(S^2) = b_0(S^2) - b_1(S^2) + b_2(S^2).$$

Substituting the values gives

$$\chi(S^2) = 1 - 0 + 1 = 2.$$

Therefore

$$\boxed{\chi(S^2) = 2.}$$

This is an example of a closed even-dimensional manifold whose Euler characteristic is not zero.

10.5 Example 3: The 3-sphere S^3

The 3-sphere S^3 is connected, so

$$b_0(S^3) = 1.$$

It has no independent 1-dimensional loops, so

$$b_1(S^3) = 0.$$

It also has no independent 2-dimensional homology classes, so

$$b_2(S^3) = 0.$$

Since S^3 is a closed connected 3-dimensional manifold, it has one top-dimensional fundamental class:

$$b_3(S^3) = 1.$$

Thus

$$b_0(S^3) = 1, \quad b_1(S^3) = 0, \quad b_2(S^3) = 0, \quad b_3(S^3) = 1.$$

Now compute the Euler characteristic:

$$\chi(S^3) = b_0(S^3) - b_1(S^3) + b_2(S^3) - b_3(S^3).$$

Substituting the Betti numbers gives

$$\chi(S^3) = 1 - 0 + 0 - 1 = 0.$$

Hence

$$\boxed{\chi(S^3) = 0.}$$

This agrees with the general fact that every closed odd-dimensional manifold has Euler characteristic zero.

10.6 Example 4: The 3-torus T^3

The 3-torus is the product

$$T^3 = S^1 \times S^1 \times S^1.$$

It is a closed 3-dimensional manifold. It has three independent circular directions. Therefore

$$b_1(T^3) = 3.$$

The Betti numbers of T^3 can be computed using the Künneth formula. A compact way to remember them is to use the polynomial

$$(1 + t)^3.$$

Expanding gives

$$(1 + t)^3 = 1 + 3t + 3t^2 + t^3.$$

The coefficients give the Betti numbers:

$$b_0(T^3) = 1, \quad b_1(T^3) = 3, \quad b_2(T^3) = 3, \quad b_3(T^3) = 1.$$

Now compute:

$$\chi(T^3) = b_0(T^3) - b_1(T^3) + b_2(T^3) - b_3(T^3).$$

Substituting the values gives

$$\chi(T^3) = 1 - 3 + 3 - 1 = 0.$$

Therefore

$$\boxed{\chi(T^3) = 0.}$$

Again, the Euler characteristic is zero because T^3 is a closed odd-dimensional manifold.

10.7 Example 5: The interval $[0, 1]$

The interval $[0, 1]$ is connected, so

$$b_0([0, 1]) = 1.$$

It has no loop, hence

$$b_1([0, 1]) = 0.$$

There are no higher-dimensional holes:

$$b_i([0, 1]) = 0 \quad \text{for } i \geq 2.$$

Therefore

$$\chi([0, 1]) = b_0([0, 1]) - b_1([0, 1]).$$

Substituting gives

$$\chi([0, 1]) = 1 - 0 = 1.$$

Hence

$$\boxed{\chi([0, 1]) = 1.}$$

Notice that $[0, 1]$ is compact, but it has boundary. Therefore it is not a closed manifold in the sense of compact manifold without boundary.

10.8 Example 6: The disk D^2

The disk D^2 is connected, so

$$b_0(D^2) = 1.$$

Every loop in the disk can be contracted to a point, so

$$b_1(D^2) = 0.$$

Although the disk is 2-dimensional, it is filled in and has boundary. It does not give a nontrivial 2-dimensional homology class. Therefore

$$b_2(D^2) = 0.$$

Thus

$$b_0(D^2) = 1, \quad b_1(D^2) = 0, \quad b_2(D^2) = 0.$$

Now compute:

$$\chi(D^2) = b_0(D^2) - b_1(D^2) + b_2(D^2).$$

Substituting gives

$$\chi(D^2) = 1 - 0 + 0 = 1.$$

Hence

$$\boxed{\chi(D^2) = 1.}$$

The disk is not closed as a manifold without boundary, because

$$\partial D^2 = S^1.$$

10.9 General rule for spheres

For the n -sphere S^n , the Betti numbers are

$$b_i(S^n) = \begin{cases} 1, & i = 0, \\ 1, & i = n, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore

$$\chi(S^n) = b_0(S^n) + (-1)^n b_n(S^n).$$

Since

$$b_0(S^n) = 1 \quad \text{and} \quad b_n(S^n) = 1,$$

we get

$$\chi(S^n) = 1 + (-1)^n.$$

Hence

$$\chi(S^n) = \begin{cases} 2, & n \text{ even,} \\ 0, & n \text{ odd.} \end{cases}$$

For example,

$$\chi(S^1) = 0, \quad \chi(S^2) = 2, \quad \chi(S^3) = 0, \quad \chi(S^4) = 2.$$

10.10 Summary table

X	Betti numbers	$\chi(X)$	Comment
S^1	1, 1	0	closed odd-dimensional
S^2	1, 0, 1	2	closed even-dimensional
S^3	1, 0, 0, 1	0	closed odd-dimensional
T^3	1, 3, 3, 1	0	closed odd-dimensional
$[0, 1]$	1, 0	1	has boundary
D^2	1, 0, 0	1	has boundary

10.11 Main conclusion

For a closed n -dimensional manifold M , Poincaré duality gives

$$b_i(M) = b_{n-i}(M).$$

If n is odd, the terms in the Euler characteristic pair with opposite signs:

$$(-1)^i b_i(M) + (-1)^{n-i} b_{n-i}(M) = 0.$$

Therefore

$$\boxed{\chi(M) = 0.}$$

Thus every closed odd-dimensional manifold has Euler characteristic zero.